H II Regions

When hot, massive stars reach the ZAMS with O or B spectral types, they do so shrouded in a cloak of gas and dust. The bulk of their radiation is emitted in the ultraviolet portion of the electromagnetic spectrum. Those photons that are produced with energies in excess of 13.6 eV can ionize the ground-state hydrogen gas (H I) in the ISM that still surrounds the newly formed star. Of course, if these **H II regions** are in equilibrium, the rate of ionization must equal the rate of recombination; photons must be absorbed and ions must be produced at the same rate that free electrons and protons recombine to form neutral hydrogen atoms. When recombination occurs, the electron does not necessarily fall directly to the ground state but can cascade downward, producing a number of lower-energy photons, many of which will be in the visible portion of the spectrum. The dominant visible wavelength photon produced in this way results from the transition between $n = 3$ and $n = 2$, the red line of the Balmer series (Hα). Consequently, because of this energy cascade, H II regions appear to fluoresce in red light.

These **emission nebulae** are considered by some to be among the most beautiful objects in the night sky. One of the more famous H II regions is the Orion nebula (M42),[14] found in the sword of the Orion constellation. M42 is part of the Orion A complex (see Fig. 12.13), which also contains a giant molecular cloud (OMC 1) and a very young cluster of stars (the Trapezium cluster). The first protostar candidates were discovered in this region as well.

The size of an H II region can be estimated by considering the requirement of equilibrium. Let N be the number of photons *per second* produced by the O or B star with sufficient energy to ionize hydrogen from the ground state ($\lambda < 91.2$ nm). Assuming that all of the

FIGURE 12.13 The H II region in Orion A is associated with a young OB association, the Trapezium cluster, and a giant molecular cloud. The Orion complex is 450 pc away. (Courtesy of the National Optical Astronomy Observatories.)

[14]M42 is the entry number in the well-known Messier catalog, a popular collection of observing objects for amateur astronomers. The Messier catalog is listed in Appendix H.

energetic photons are ultimately absorbed by the hydrogen in the H II region, the rate of photon creation must equal the rate of recombination. If this equilibrium condition did not develop, the size of the region would continue to grow as the photons traveled ever farther before encountering un-ionized gas.

Next, let $\alpha n_e n_H$ be the number of recombinations *per unit volume per second*, where α is a quantum-mechanical recombination coefficient that describes the likelihood that an electron and a proton can form a hydrogen atom, given their number densities (obviously, the more electrons and protons that are present, the greater the chance of recombination; hence the product $n_e n_H$).[15] At about 8000 K, a temperature characteristic of H II regions, $\alpha = 3.1 \times 10^{-19}$ m^3 s^{-1}. If we assume that the gas is composed entirely of hydrogen and is electrically neutral, then for every ion produced, one electron must have been liberated, or $n_e = n_H$. With this equality, the expression for the recombination rate can be multiplied by the volume of the H II region, assumed here to be spherical, and then set equal to the number of ionizing photons produced per second. Finally, solving for the radius of the H II region gives

$$\boxed{r_S \simeq \left(\frac{3N}{4\pi\alpha}\right)^{1/3} n_H^{-2/3}.} \tag{12.32}$$

r_S is called the **Strömgren radius**, after Bengt Strömgren (1908–1987), the astrophysicist who first carried out the analysis in the late 1930s.

Example 12.3.1. From Appendix G, the effective temperature and luminosity of an O6 star are $T_e \simeq 45{,}000$ K and $L \simeq 1.3 \times 10^5\ \mathrm{L}_\odot$, respectively. According to Wien's law (Eq. 3.15), the peak wavelength of the blackbody spectrum is given by

$$\lambda_{\max} = \frac{0.0029 \text{ m K}}{T_e} = 64 \text{ nm}.$$

Since this is significantly shorter than the 91.2-nm limit necessary to produce ionization from the hydrogen ground state, it can be assumed that most of the photons created by an O6 star are capable of causing ionization.

The energy of one 64-nm photon can be calculated from Eq. (5.3), giving

$$E_\gamma = \frac{hc}{\lambda} = 19 \text{ eV}.$$

Now, assuming for simplicity that all of the emitted photons have the same (peak) wavelength, the total number of photons produced by the star per second is just

$$N \simeq L/E_\gamma \simeq 1.6 \times 10^{49} \text{ photons s}^{-1}.$$

Lastly, taking $n_H \sim 10^8$ m^{-3} to be a typical value an H II region, we find

$$r_S \simeq 3.5 \text{ pc}.$$

Values of r_S range from less than 0.1 pc to greater than 100 pc.

---

[15]Note that this expression is somewhat analogous to Eq. (10.29), the generalized nuclear reaction rate equation.

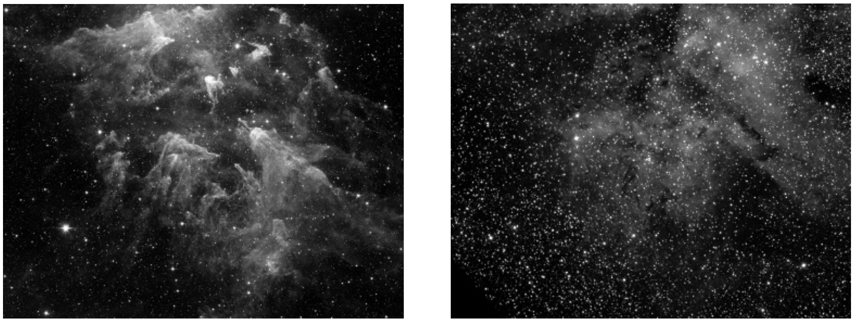
The Effects of Massive Stars on Gas Clouds

As a massive star forms, the protostar will initially appear as an infrared source embedded inside the molecular cloud. With the rising temperature, first the dust will vaporize, then the molecules will dissociate, and finally, as the star reaches the main sequence, the gas immediately surrounding it will ionize, resulting in the creation of an H II region inside of an existing H I region.

Now, because of the star's high luminosity, radiation pressure will begin to drive significant amounts of mass loss, which then tends to disperse the remainder of the cloud. If several O and B stars form at the same time, it may be that much of the mass that has not yet become gravitationally bound to more slowly forming low-mass protostars will be driven away, halting any further star formation. Moreover, if the cloud was originally marginally bound (near the limit of criticality), the loss of mass will diminish the potential energy term in the virial theorem, with the result that the newly formed cluster of stars and protostars will become unbound (i.e., the stars will tend to drift apart). Figure 12.14 shows such a process under way in the Carina Nebula, located approximately 3000 pc from Earth. Another famous example of the effects of ionizing radiation of nearby massive stars is the production of the pillars in M16, the Eagle Nebula (Fig. 12.15).

OB Associations

Groups of stars that are dominated by O and B main-sequence stars are referred to as **OB associations**. Studies of their individual kinematic velocities and masses generally lead to the conclusion that they cannot remain gravitationally bound to one another as permanent stellar clusters. One such example is the Trapezium cluster in the Orion A complex, believed to be less than 10 million years old. It is currently densely populated with stars



(a) (b)

FIGURE 12.14 (a) An infrared image of a portion of the Carina Nebula. Eta Carina, a very young and marginally stable star of more than 100 $M_\odot$ is located above the image. The strong winds and intense ultraviolet radiation from Eta Carina and other massive stars in the region are shredding the nebula. Other, lower-mass newborn stars, such as those just above the pillar to the right of center in the image, are being inhibited from growing larger because of the destruction of the nebula by their much more massive siblings. [NASA/JPL-Caltech/N. Smith (University of Colorado at Boulder)] (b) The same region observed in visible light. Much less detail is observable because of the obscuration due to dust in the cloud. (NOAO)

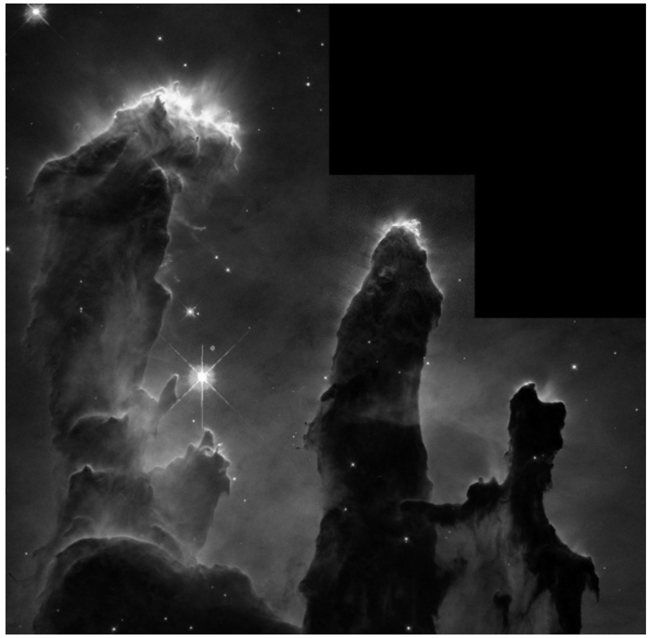


FIGURE 12.15 The giant gas pillars of the Eagle Nebula (M16). The left-most pillar is more than 1 pc long from base to top. Ionizing radiation from massive newborn stars off the top edge of the image are causing the gas in the cloud to photoevaporate. [Courtesy of NASA, ESA, STScI, J. Hester and P. Scowen (Arizona State University).]

($> 2 \times 10^3$ pc^{-3}), most of which have masses in the range of 0.5 to 2.0 $M_\odot$. Doppler shift measurements of the radial velocities of ^{13}CO show that the gas in the vicinity is very turbulent. Apparently, the nearby O and B stars are dispersing the gas, and the cluster is becoming unbound.

T Tauri Stars

T Tauri stars are an important class of low-mass pre-main-sequence objects that represent a transition between stars that are still shrouded in dust (IR sources) and main-sequence stars. T Tauri stars, named after the first star of their class to be identified (located in the constellation of Taurus), are characterized by unusual spectral features and by large and fairly rapid irregular variations in luminosity, with timescales on the order of days. The positions of T Tauri stars on the H–R diagram are shown in Fig. 12.16; theoretical pre-main-sequence evolutionary tracks are also included. The masses of T Tauri stars range from 0.5 to about 2 $M_\odot$.

Many T Tauri stars exhibit strong emission lines from hydrogen (the Balmer series), from Ca II (the H and K lines), and from iron, as well as absorption lines of lithium. Interestingly, forbidden lines of [O I] and [S II] are also present in the spectra of many T Tauri stars.[16] The

[16]Recall the discussion of allowed and forbidden transitions in the Sun's corona (Section 11.2).

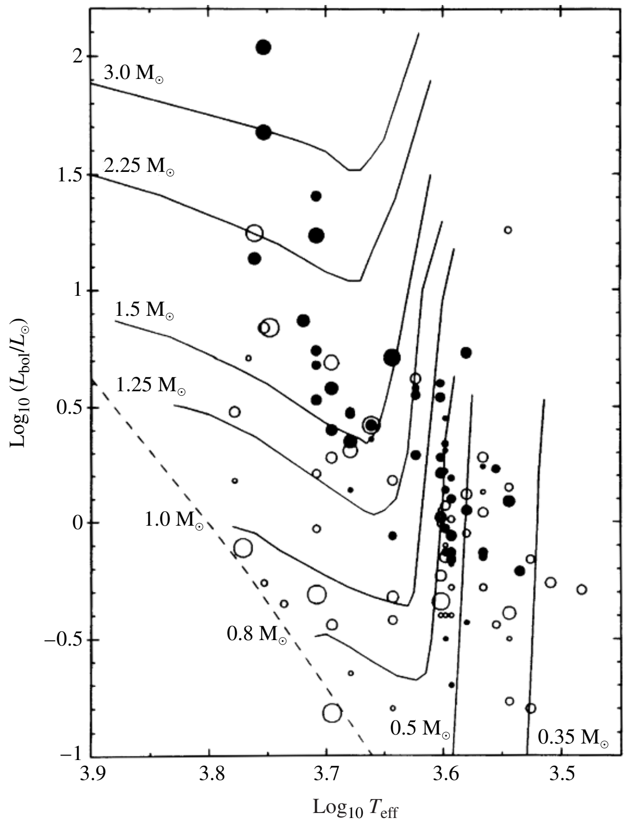


FIGURE 12.16 The positions of T Tauri stars on the H–R diagram. The sizes of the circles indicate the rate of rotation. Stars with strong emission lines are indicated by filled circles, and weak emission line stars are represented by open circles. Theoretical pre-main-sequence evolutionary tracks are also included. (Figure adapted from Bertout, *Annu. Rev. Astron. Astrophys.*, *27*, 351, 1989. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 27, ©1989 by Annual Reviews Inc.)

existence of forbidden lines in a spectrum is an indication of extremely low gas densities. (Note that, to distinguish them from "allowed" lines, forbidden lines are usually indicated by square brackets, e.g., [O I].)

Not only can information be gleaned from spectra by determining which lines are present and with what strengths, but information is also contained in the *shapes* of those lines as a function of wavelength.[17] An important example is found in the shapes of some of the lines in T Tauri stars. The Hα line often exhibits the characteristic shape shown in Fig. 12.17(a). Superimposed on a rather broad emission peak is an absorption trough at the short-wavelength edge of the line. This unique line shape is known as a **P Cygni profile**, after the first star observed to have emission lines with blueshifted absorption components.

[17]Line profiles were first discussed in Section 9.5; recall also Fig. 12.10.

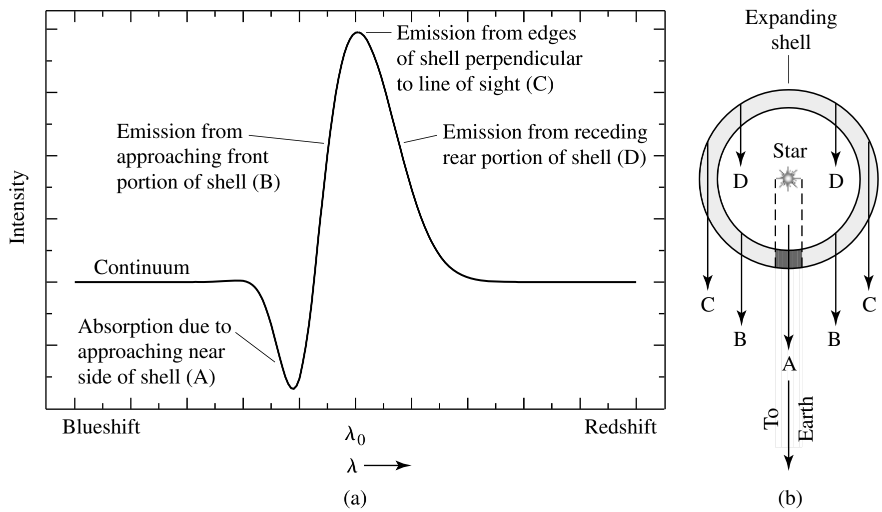


FIGURE 12.17 (a) A spectral line exhibiting a P Cygni profile is characterized by a broad emission peak with a superimposed blueshifted absorption trough. (b) A P Cygni profile is produced by an expanding mass shell. The emission peak is due to the outward movement of material perpendicular to the line of sight, whereas the blueshifted absorption feature is caused by the approaching matter in the shaded region, intercepting photons coming from the central star.

The interpretation given for the existence of P Cygni profiles in a star's spectrum is that the star is experiencing significant mass loss. Recall from Kirchhoff's laws (Section 5.1) that emission lines are produced by a hot, diffuse gas when there is little intervening material between the source and the observer. In this case the emission source is that portion of the expanding shell of the T Tauri star that is moving nearly perpendicular to the line of sight, as illustrated by the geometry shown in Fig. 12.17(b). Absorption lines are the result of light passing through a cooler, diffuse gas; the shaded portion of the expanding shell absorbs the photons emitted by the hotter star behind it. Since the shaded part of the shell (A) is moving toward the observer, the absorption is blueshifted relative to the emission component (typically by 80 km s^{-1} for T Tauri stars). The mass loss rates of T Tauri stars average about $\dot{M} = 10^{-8}\ \mathrm{M}_\odot\ \mathrm{yr}^{-1}$.[18]

In some extreme cases, line profiles of T Tauri stars have gone from P Cygni profiles to *inverse* P Cygni profiles (redshifted absorption) on timescales of days, indicating mass accretion rather than mass loss. Mass accretion rates appear to be on the same order as mass loss rates. Apparently the environment around a T Tauri star is very unstable.

FU Orionis Stars

In some instances, it appears that T Tauri stars have gone through very significant increases in mass accretion rates, reaching values on the order of $\dot{M} = 10^{-4}\ \mathrm{M}_\odot\ \mathrm{yr}^{-1}$. At the same time the luminosities of the stars increase by four magnitudes or more, with the increases lasting for decades. The first star observed to undergo this abrupt increase in accretion

[18]This value is much higher than the Sun's current rate of mass loss ($10^{-14}\ \mathrm{M}_\odot\ \mathrm{yr}^{-1}$; see Example 11.2.1).

was FU Orionis, for which the **FU Orionis stars** are named. Apparently, instabilities in a circumstellar accretion disk around an FU Orionis star can result in on the order of 0.01 $M_\odot$ being dumped onto the central star over the century or so duration of the outburst. During that time the inner disk can outshine the central star by a factor of 100 to 1000, while strong, high-velocity winds in excess of 300 km s^{-1} occur. It has been suggested that T Tauri stars may go through several FU Orionis events during their lifetimes.

Herbig Ae/Be Stars

Closely related to the T-Tauri stars are **Herbig Ae/Be stars**, named for George Herbig. These pre-main-sequence stars are of spectral types A or B and have strong emission lines (hence the Ae/Be designations). Their masses range from 2 to 10 $M_\odot$ and they tend to be enveloped in some remaining dust and gas. He Ae/Be stars are not as thoroughly studied as T-Tauri stars, in large part because of their much shorter lifetimes (recall Table 12.1) and in part because fewer intermediate-mass than lower-mass stars form from a cloud (Fig. 12.12).

Herbig–Haro Objects

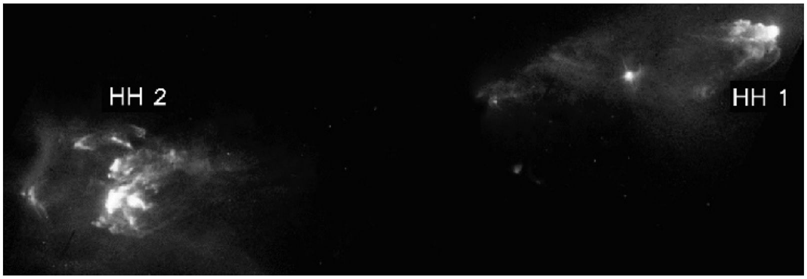
Along with expanding shells, mass loss during pre-main-sequence evolution can also occur from **jets** of gas that are ejected in narrow beams in opposite directions.[19] **Herbig–Haro objects**, first discovered in the vicinity of the Orion nebula in the early 1950s by George Herbig and Guillermo Haro (1913–1988), are apparently associated with the jets produced by young protostars, such as T Tauri stars. As the jets expand supersonically into the interstellar medium, collisions excite the gas, resulting in bright objects with emission-line spectra. Figure 12.18(a) shows a Hubble Space Telescope image of the Herbig–Haro objects HH 1 and HH 2, which were created by material ejected at speeds of several hundred kilometers per second from a star shrouded in a cocoon of dust. The jets associated with another Herbig–Haro object, HH 47, are shown in Fig. 12.18(b).

Continuous emission is also observed in some protostellar objects and is due to the reflection of light from the parent star. A circumstellar accretion disk is apparent in Fig. 12.19 around HH 30. The surfaces of the disk are illuminated by the central star, which is again hidden from view behind the dust in the disk. Also apparent are jets originating from deep within the accretion disk, possibly from the central star itself. These accretion disks seem to be responsible for many of the characteristics associated with the protostellar objects, including emission lines, mass loss, jets, and perhaps even some of the luminosity variations. Unfortunately, details concerning the physical processes involved are not fully understood. An early model of the production of Herbig–Haro objects like HH 1 and HH 2 is shown in Fig. 12.20.

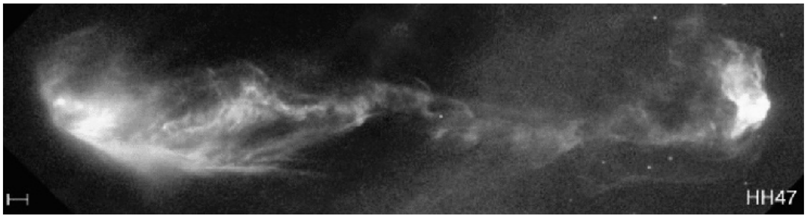
Young Stars with Circumstellar Disks

Observations have revealed that other young stars also possess circumstellar disks of material orbiting them. Two well-known examples are Vega and β Pictoris. An infrared image of β Pic and its disk is shown in Fig. 12.21. β Pic has also been observed in the ultraviolet lines of Fe II by the Hubble Space Telescope. It appears that clumps of material are falling from

[19] As we shall see in later chapters, astrophysical jets occur in a variety of phenomena over enormous ranges of energy and physical size.



(a)



(b)

FIGURE 12.18 (a) The Herbig–Haro objects HH 1 and HH 2 are located just south of the Orion nebula and are moving away from a young protostar hidden inside a dust cloud near the center of the image. [Courtesy of J. Hester (Arizona State University), the WF/PC 2 Investigation Definition Team, and NASA.] (b) A jet associated with HH 47. The scale at the lower left is 1000 AU. (Courtesy of J. Morse/STScI, and NASA.)

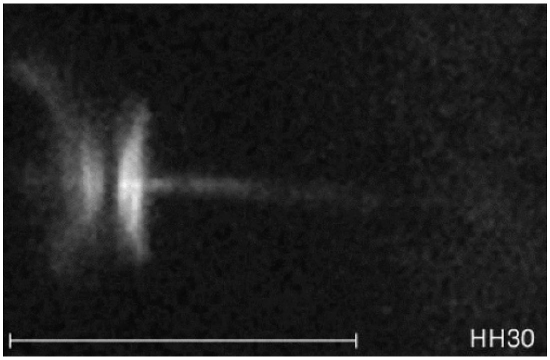


FIGURE 12.19 The circumstellar disk and jets of the protostellar object, HH 30. The central star is obscured by dust in the plane of the disk. The scale at the lower left is 1000 AU. [Courtesy of C. Burrows (STScI and ESA), the WF/PC 2 Investigation Definition Team, and NASA.]

the disk into the star at the rate of two or three per week. Larger objects may be forming in the disk as well, possibly protoplanets. It has been suggested that these disks may in fact be **debris disks** rather than accretion disks, meaning that the observed material is due to collisions between objects already formed in the disks. An artist's conception of the β Pic system is shown in Fig. 12.22.

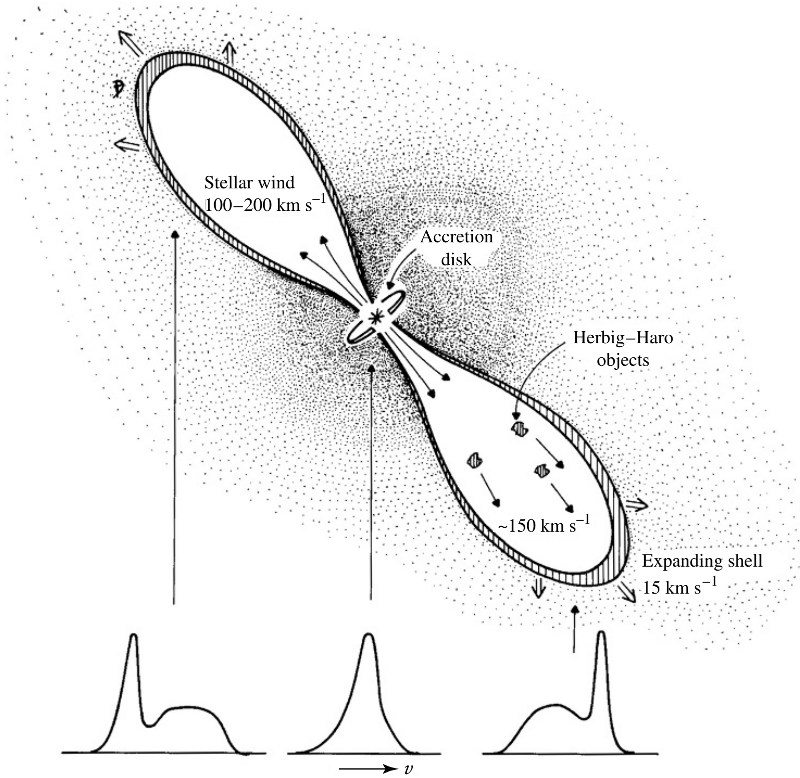


FIGURE 12.20 An early model of a T Tauri star with an accretion disk. The disk powers and collimates jets that expand into the interstellar medium, producing Herbig–Haro objects. (Figure adapted from Snell, Loren, and Plambeck, *Ap. J. Lett.*, *239*, L17, 1980.)

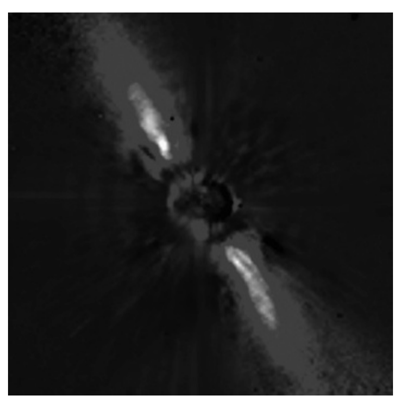


FIGURE 12.21 An infrared image of β Pictoris, showing its circumstellar debris disk. (European Southern Observatory)

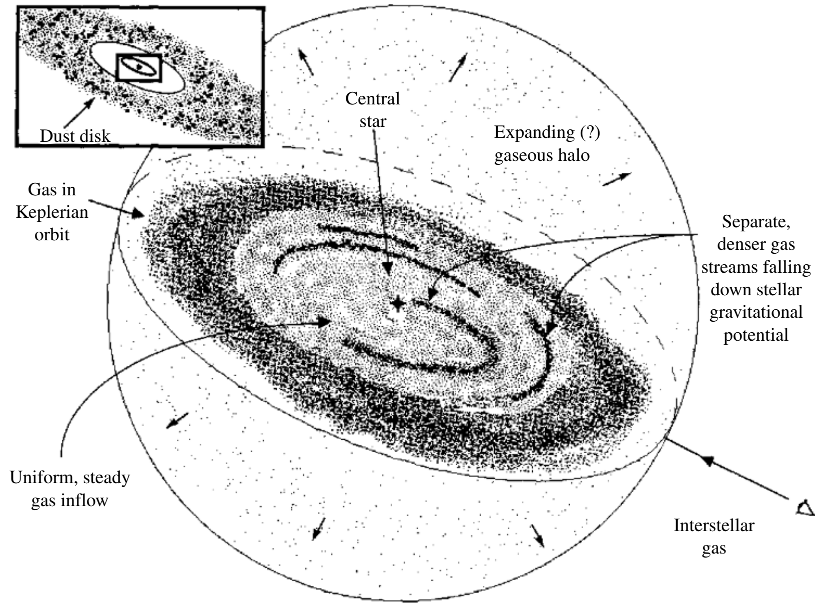


FIGURE 12.22 An artist's conception of the β Pictoris system. Clumps of material appear to be falling into the star at the rate of two or three clumps per week. Some matter may also be leaving the system as an expanding halo. (Figure adapted from Boggess et al., *Ap. J. Lett.*, *377*, L49, 1991.)

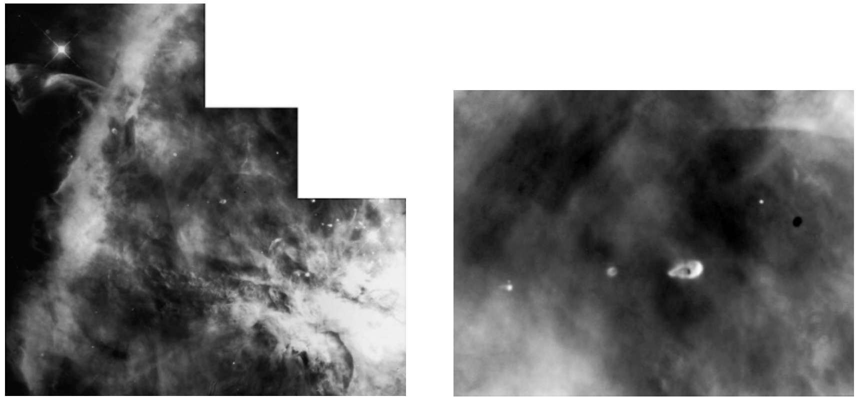
Proplyds

Shortly after the December 1993 refurbishment mission of the Hubble Space Telescope, HST made observations of the Orion Nebula. The images in Fig. 12.23 were obtained using the emission lines of Hα, [N II], and [O III]. Analysis of the data has revealed that 56 of the 110 stars brighter than $V = 21$ mag are surrounded by disks of circumstellar dust and gas. The circumstellar disks, termed **proplyds**, appear to be protoplanetary disks associated with young stars that are less than 1 million years old. Based on observations of the ionized material in the proplyds, the disks seem to have masses much greater than 2×10^{25} kg (for reference, the mass of Earth is 5.974×10^{24} kg).

Circumstellar Disk Formation

Apparently, disk formation is fairly common during the collapse of protostellar clouds. Undoubtedly this is due to the spin-up of the cloud as required by the conservation of angular momentum. As the radius of the protostar decreases, so does its moment of inertia. This implies that in the absence of external torques, the protostar's angular velocity must increase. It is left as an exercise (Problem 12.18) to show that by including a centripetal acceleration term in Eq. (12.19) and requiring conservation of angular momentum, the collapse perpendicular to the axis of rotation can be halted before the collapse along the axis, resulting in disk formation.

A problem immediately arises when the effect of angular momentum is included in the collapse. Conservation of angular momentum arguments lead us to expect that all main-



(a) (b)

FIGURE 12.23 Images of the Orion Nebula (M42) obtained by the Hubble Space Telescope. Note that (b) is an enlarged view of the central region of (a). Numerous proplyds are visible in the field of view of the camera. (Courtesy of C. Robert O'Dell/Vanderbilt University, NASA, and ESA.)

sequence stars ought to be rotating very rapidly, at rates close to breakup. However, observations show that this is not generally the case. Apparently the angular momentum is transferred away from the collapsing star. One suggestion (discussed in Section 11.2) is that magnetic fields, anchored to convection zones within the stars and coupled to ionized stellar winds, slow the rotation by applying torques. Evidence in support of this idea exists in the form of apparent solar-like coronal activity in the outer atmospheres of many T Tauri stars.

Along with the problems associated with rotation and magnetic fields, mass loss may also play an important role in the evolution of pre-main-sequence stars. Although these problems are being investigated, much work remains to be done before we can hope to understand all of the details of protostellar collapse and pre-main-sequence evolution.

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PROBLEMS

12.1 In a certain part of the North American Nebula, the amount of interstellar extinction in the visual wavelength band is 1.1 magnitudes. The thickness of the nebula is estimated to be 20 pc, and it is located 700 pc from Earth. Suppose that a B spectral class main-sequence star is observed in the direction of the nebula and that the absolute visual magnitude of the star is known to be $M_V = -1.1$ from spectroscopic data. Neglect any other sources of extinction between the observer and the nebula.

(a) Find the apparent visual magnitude of the star if it is lying just in front of the nebula.

(b) Find the apparent visual magnitude of the star if it is lying just behind the nebula.

(c) Without taking the existence of the nebula into consideration, based on its apparent magnitude, how far away does the star in part (b) appear to be? What would be the percentage error in determining the distance if interstellar extinction were neglected?

12.2 Estimate the temperature of a dust grain that is located 100 AU from a newly formed F0 main-sequence star. *Hint:* Assume that the dust grain is in thermal equilibrium—meaning that the amount of energy absorbed by the grain in a given time interval must equal the amount of energy radiated away during the same interval of time. Assume also that the dust grain is spherically symmetric and emits and absorbs radiation as a perfect blackbody. You may want to refer to Appendix G for the effective temperature and radius of an F0 main-sequence star.

12.3 The Boltzmann factor, $e^{-(E_2-E_1)/kT}$, helps determine the relative populations of energy levels (see Section 8.1). Using the Boltzmann factor, estimate the temperature required for a hydrogen atom's electron and proton to go from being anti-aligned to being aligned. Are the temperatures in H I clouds sufficient to produce this low-energy excited state?

12.4 An H I cloud produces a 21-cm line with an optical depth at its center of $\tau_H = 0.5$ (the line is optically thin). The temperature of the gas is 100 K, the line's full width at half-maximum is 10 km s^{-1}, and the average atomic number density of the cloud is estimated to be 10^7 m^{-3}. From this information and Eq. (12.7), find the thickness of the cloud. Express your answer in pc.

12.5 Using an approach analogous to the development of Eq. (10.29) for nuclear reaction rates, make a crude estimate of the number of random collisions per cubic meter per second between CO and H_2 molecules in a giant molecular cloud that has a temperature of 15 K and a number density of $n_{H_2} = 10^8$ m^{-3}. Assume (incorrectly) that the molecules are spherical in shape with radii of approximately 0.1 nm, the characteristic size of an atom.

12.6 Explain why astronomers would use the isotopomers ^{13}CO or C^{18}O, rather than the more common CO molecule, to probe deeply into a giant molecular cloud.

12.7 The rotational kinetic energy of a molecule is given by

$$E_{\rm rot} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I},$$

where L is the molecule's angular momentum and I is its moment of inertia. The angular momentum is restricted by quantum mechanics to the discrete values

$$L = \sqrt{\ell(\ell+1)}\hbar$$

where $\ell = 0, 1, 2, \ldots$.

(a) For a diatomic molecule,

$$I = m_1 r_1^2 + m_2 r_2^2,$$

where m_1 and m_2 are the masses of the individual atoms and r_1 and r_2 are their separations from the center of mass of the molecule. Using the ideas developed in Section 2.3, show that I may be written as

$$I = \mu r^2,$$

where μ is the reduced mass and r is the separation between the atoms in the molecule.

(b) The separation between the carbon and oxygen atoms in CO is approximately 0.12 nm, and the atomic masses of ^{12}C, ^{13}C, and ^{16}O are 12.000 u, 13.003 u, and 15.995 u, respectively. Calculate the moments of inertia for ^{12}CO and ^{13}CO.

(c) What is the wavelength of the photon that is emitted by ^{12}CO during a transition between the rotational angular momentum states $\ell = 3$ and $\ell = 2$? To which part of the electromagnetic spectrum does this correspond?

(d) Repeat part (c) for ^{13}CO. How do astronomers distinguish among different isotopes in the interstellar medium?

12.8 **(a)** Equations (12.10) and (12.11) illustrate a cooling mechanism for a molecular cloud accomplished through the excitation of oxygen atoms. Explain why the excitation of hydrogen rather than oxygen is not an effective cooling mechanism.

(b) Why are the temperatures of hot cores significantly greater than dense cores?

12.9 In light of the cooling mechanisms discussed for molecular clouds, explain why dense cores are generally cooler than the surrounding giant molecular clouds, and why GMCs are cooler than diffuse molecular clouds.

12.10 Calculate the Jeans length for the dense core of the giant molecular cloud in Example 12.2.1.

12.11 Show that the Jeans mass (Eq. 12.14) can also be written in the form

$$M_J = \frac{c_J v_T^4}{P_0^{1/2} G^{3/2}} \tag{12.33}$$

where the isothermal sound speed, v_T, is given by Eq. (12.18), P_0 is the pressure associated with the density ρ_0 and temperature T, and $c_J \simeq 5.46$ is a dimensionless constant.

12.12 By invoking the requirements of hydrostatic equilibrium, explain why the assumption of a constant gas pressure P_0 in Eq. (12.33) cannot be correct for a static cloud without magnetic fields. What does that imply about the assumptions of constant mass density in an isothermal molecular cloud having a constant composition throughout?

12.13 **(a)** By using the ideal gas law, calculate $|dP/dr| \approx |\Delta P/\Delta r| \sim P_c/R_J$ at the beginning of the collapse of the dense core of a giant molecular cloud, where P_c is an approximate value for the central pressure of the cloud. Assume that $P = 0$ at the edge of the molecular cloud and take its mass and radius to be the Jeans values found in Example 12.2.1 and in Problem 12.10. You should also assume the cloud temperature and density given in Example 12.2.1.

(b) Show that, given the accuracy of our crude estimates, $|dP/dr|$ found in part (a) is much smaller than $GM_r\rho/r^2$. What does this say about the core's dynamics?

(c) Show that as long as the collapse remains isothermal, the contribution of dP/dr in Eq. (10.5) continues to decrease relative to $GM_r\rho/r^2$, supporting the assumption made in Eq. (12.19) that dP/dr can be neglected once free-fall collapse begins.

12.14 Assuming that the free-fall acceleration of the surface of a collapsing cloud remains constant during the entire collapse, derive an expression for the free-fall time. Show that your answer differs from Eq. (12.26) only by a term of order unity.

12.15 Using Eq. (10.84), estimate the adiabatic sound speed of the dense core of the giant molecular cloud discussed in Examples 12.2.1 and 12.2.2. Use this speed to find the amount of time required for a sound wave to cross the cloud, $t_s = 2R_J/v_s$, and compare your answer to the estimate of the free-fall time found in Example 12.2.2. Explain your result.

12.16 Using the information contained in the text, derive Eq. (12.28).

12.17 Estimate the gravitational energy per unit volume in the dense core of the giant molecular cloud in Example 12.2.1, and compare that with the magnetic energy density that would be contained in the cloud if it had a magnetic field of uniform strength, $B = 1$ nT. [*Hint:* Refer to Eq. (11.9).] Could magnetic fields play a significant role in the collapse of a cloud?

12.18 **(a)** Beginning with Eq. (12.19), adding a centripetal acceleration term, and using conservation of angular momentum, show that the collapse of a cloud will stop in the plane perpendicular to its axis of rotation when the radius reaches

$$r_f = \frac{\omega_0^2 r_0^4}{2GM_r}$$

where M_r is the interior mass, and ω_0 and r_0 are the original angular velocity and radius of the surface of the cloud, respectively. Assume that the initial radial velocity of the cloud is zero and that $r_f \ll r_0$. You may also assume (incorrectly) that the cloud rotates as a rigid body during the entire collapse. *Hint:* Recall from the discussion leading to Eq. (11.7) that $d^2r/dt^2 = v_r\, dv_r/dr$. (Since no centripetal acceleration term exists for collapse along the rotation axis, disk formation is a consequence of the original angular momentum of the cloud.)

(b) Assume that the original cloud had a mass of 1 $M_\odot$ and an initial radius of 0.5 pc. If collapse is halted at approximately 100 AU, find the initial angular velocity of the cloud.

(c) What was the original rotational velocity (in m s^{-1}) of the edge of the cloud?

(d) Assuming that the moment of inertia is approximately that of a uniform solid sphere, $I_{\text{sphere}} = \frac{2}{5}Mr^2$, when the collapse begins and that of a uniform disk, $I_{\text{disk}} = \frac{1}{2}Mr^2$, when it stops, determine the rotational velocity at 100 AU.

(e) Calculate the time required, after the collapse has stopped, for a piece of mass to make one complete revolution around the central protostar. Compare your answer with the orbital period at 100 AU expected from Kepler's third law. Why would you not expect the two periods to be identical?

12.19 Assuming a mass loss rate of 10^{-7} $M_\odot$ yr^{-1} and a stellar wind velocity of 80 km s^{-1} from a T Tauri star, estimate the mass density of the wind at a distance of 100 AU from the star. (*Hint:* Refer to Example 11.2.1.) Compare your answer with the density of the giant molecular cloud in Example 12.2.1.

CHAPTER

13

Main Sequence and Post-Main-Sequence Stellar Evolution

13.1 ■ EVOLUTION ON THE MAIN SEQUENCE

In Section 10.6 we learned that the existence of the main sequence is due to the nuclear reactions that convert hydrogen into helium in the cores of stars. The evolutionary process of protostellar collapse to the zero-age main sequence was discussed in Chapter 12. In this chapter we will follow the lives of stars as they age, beginning on the main sequence. This evolutionary process is an inevitable consequence of the relentless force of gravity and the change in chemical composition due to nuclear reactions.

Stellar Evolution Timescales

To maintain their luminosities, stars must tap sources of energy contained within, either nuclear or gravitational.[1] Pre-main-sequence evolution is characterized by two basic timescales: the free-fall timescale (Eq. 12.26) and the thermal Kelvin–Helmholtz timescale (Eq. 10.24). Main-sequence and post-main-sequence evolution are also governed by a third timescale, the timescale of nuclear reactions (Eq. 10.25). As we saw in Example 10.3.2, the nuclear timescale is on the order of 10^{10} years for the Sun, much longer than the Kelvin–Helmholtz timescale of roughly 10^7 years, estimated in Example 10.3.1. It is the difference in timescales for the various phases of evolution of individual stars that explains why approximately 80% to 90% of all stars in the solar neighborhood are observed to be main-sequence stars (see Section 8.2); we are more likely to find stars on the main sequence simply because that stage of evolution requires the most time; later stages of evolution proceed more rapidly. However, as a star switches from one nuclear source to the next, gravitational energy can play a major role and the Kelvin–Helmholtz timescale will again become important.

[1]We have already seen in Problem 10.3 that chemical energy cannot play a significant role in the energy budgets of stars.

Width of the Main Sequence

Careful study of the main sequence of an observational H–R diagram such as Fig. 8.13 or the observational mass–luminosity relation (Fig. 7.7) reveals that these curves are not simply thin lines but have finite widths. The widths of the main sequence and the mass–luminosity relation are due to a number of factors, including observational errors, differing chemical compositions of the individual stars in the study, and varying stages of evolution on the main sequence.

Low-Mass Main-Sequence Evolution

In this section, we will consider the evolution of stars on the main sequence. Although all stars on the main sequence are converting hydrogen into helium and, as a result, share similar evolutionary characteristics, differences do exist. For instance, as was mentioned in Section 10.6, zero-age main-sequence (ZAMS) stars with masses greater than about 1.2 $M_\odot$ have convective cores due to the highly temperature-dependent CNO cycle. On the other hand, ZAMS stars with masses less than 1.2 $M_\odot$ are dominated by the less temperature-dependent pp chain. This implies that ZAMS stars in the range 0.3 $M_\odot$ to 1.2 $M_\odot$ possess radiative cores. However, the lowest-mass ZAMS stars again have convective cores because their high surface opacities drive surface convection zones deep into the interior, making the entire star convective.

First consider a typical low-mass main-sequence star such as the Sun. As we noted in the discussion of Fig. 11.1, the Sun's luminosity, radius, and temperature have all increased steadily since it reached the ZAMS 4.57 Gyr ago. This evolution occurs because, as the pp chain converts hydrogen into helium, the mean molecular weight μ of the core increases (Eq. 10.16). According to the ideal gas law (Eq. 10.11), unless the density and/or temperature of the core also increases, there will be insufficient gas pressure to support the overlying layers of the star. As a result, the core must be compressed. While the density of the core increases, gravitational potential energy is released, and, as required by the virial theorem (Section 2.4), half of the energy is radiated away and half of the energy goes into increasing the thermal energy and hence the temperature of the gas. One consequence of this temperature increase is that the region of the star that is hot enough to undergo nuclear reactions increases slightly during the main-sequence phase of evolution. In addition, since the pp chain nuclear reaction rate goes as $\rho X^2 T_6^4$ (see Eq. 10.47), the increased temperature and density more than offset the decrease in the mass fraction of hydrogen, and the luminosity of the star slowly increases, along with its radius and effective temperature.

Main-sequence and post-main-sequence evolutionary tracks of stars of various masses were first computed in a pioneering study by Icko Iben, Jr., and published in the mid-1960s. Modern calculations of theoretical evolutionary tracks that include the effects of convective overshooting as well as mass lost from stars during their lifetimes are shown in Fig. 13.1.[2] According to the calculations, the amount of time required to evolve from the zero-age main sequence to points indicated in Fig. 13.1 are as given in Table 13.1. The locus of points

[2]Convective overshooting takes into consideration the inertia of a convective bubble, which causes it to travel some distance into an otherwise radiative region of the star.

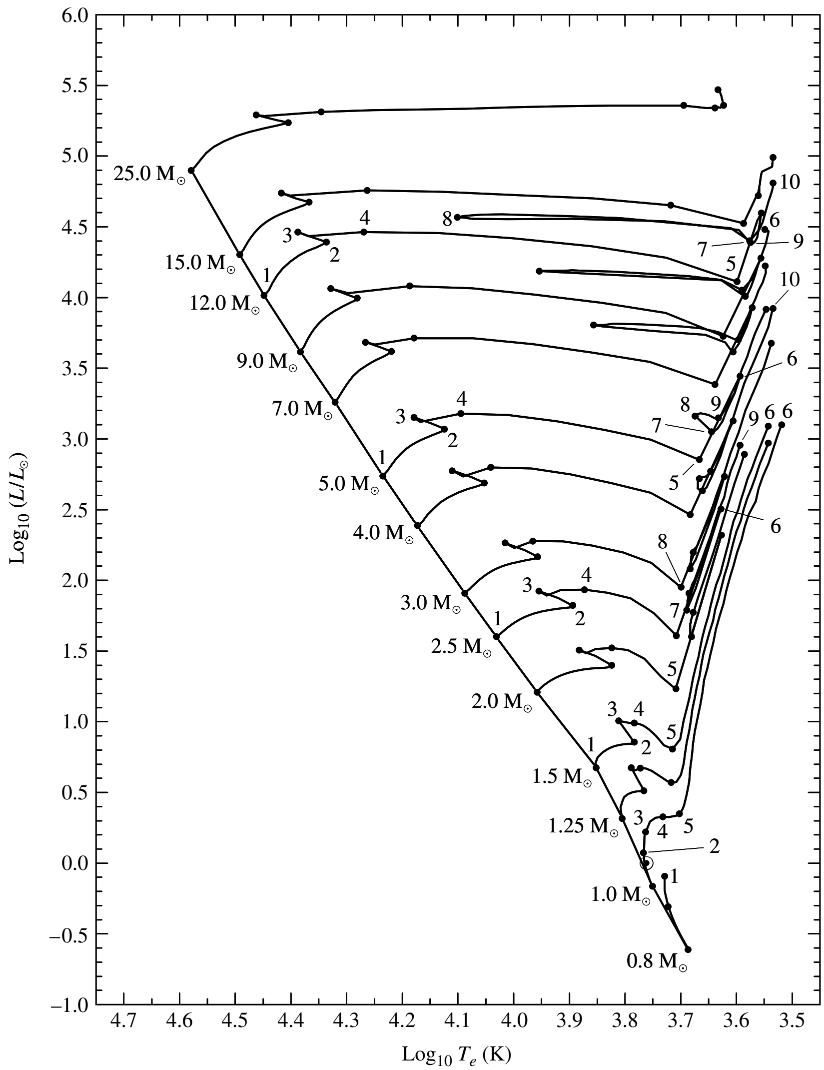


FIGURE 13.1 Main-sequence and post-main-sequence evolutionary tracks of stars with an initial composition of $X = 0.68$, $Y = 0.30$, and $Z = 0.02$. The location of the present-day Sun (see Fig. 13.2) is depicted by the solar symbol ($\odot$) between points 1 and 2 on the 1 $M_\odot$ track. The elapsed times to points indicated on the diagram are given in Table 13.1. To enhance readability, only the points on the evolutionary tracks for 0.8, 1.0, 1.5, 2.5, 5.0, and 12.0 $M_\odot$ are labeled. The model calculations include mass loss and convective overshooting. The diagonal line connecting the locus of points 1 is the zero-age main sequence. For complete, and annotated, evolutionary tracks of 1 $M_\odot$ and 5 $M_\odot$ stars, see Figs. 13.4 and 13.5, respectively. (Data from Schaller et al., *Astron. Astrophys. Suppl.*, *96*, 269, 1992.)

TABLE 13.1 The elapsed times since reaching the zero-age main sequence to the indicated points in Fig. 13.1, measured in millions of years (Myr). (Data from Schaller et al., *Astron. Astrophys. Suppl.*, *96*, 269, 1992.)

Initial Mass	1	2	3	4	5
$(M_\odot)$	6	7	8	9	10
25	0	6.33044	6.40774	6.41337	6.43767
	6.51783	7.04971	7.0591		
15	0	11.4099	11.5842	11.5986	11.6118
	11.6135	11.6991	12.7554		
12	0	15.7149	16.0176	16.0337	16.0555
	16.1150	16.4230	16.7120	17.5847	17.6749
9	0	25.9376	26.3886	26.4198	26.4580
	26.5019	27.6446	28.1330	28.9618	29.2294
7	0	42.4607	43.1880	43.2291	43.3388
	43.4304	45.3175	46.1810	47.9727	48.3916
5	0	92.9357	94.4591	94.5735	94.9218
	95.2108	99.3835	100.888	107.208	108.454
4	0	162.043	164.734	164.916	165.701
	166.362	172.38	185.435	192.198	194.284
3	0	346.240	352.503	352.792	355.018
	357.310	366.880	420.502	440.536	
2.5	0	574.337	584.916	586.165	589.786
	595.476	607.356	710.235	757.056	
2	0	1094.08	1115.94	1117.74	1129.12
	1148.10	1160.96	1379.94	1411.25	
1.5	0	2632.52	2690.39	2699.52	2756.73
	2910.76				
1.25	0	4703.20	4910.11	4933.83	5114.83
	5588.92				
1	0	7048.40	9844.57	11386.0	11635.8
	12269.8				
0.8	0	18828.9	25027.9		

labeled 1 represents the theoretical ZAMS, with the present-day Sun located between points 1 and 2 on the 1 $M_\odot$ track.

We discussed solar models in some detail in Section 11.1, where Figs. 11.3–11.7 showed the internal structure of one such model as a function of radius. In particular, Fig. 11.3 illustrated the partial depletion of hydrogen in the core, together with the accompanying increase in the amount of helium. The internal structure of the present-day Sun is also shown in Fig. 13.2, this time as a function of interior mass. Along with radius, density, temperature, pressure, and luminosity, the figure illustrates the mass fractions of the species ${}^1_1$H, ${}^3_2$He, ${}^{12}_{6}$C, ${}^{14}_{7}$N, and ${}^{16}_{8}$O. As the star's evolution on the main sequence continues, eventually the hydrogen at its center will be completely depleted. Such a situation is illustrated in Fig. 13.3 for a 1 $M_\odot$ star approximately 9.8 Gyr after arriving on the ZAMS; this model roughly corresponds to point 3 in Fig. 13.1.

With the depletion of hydrogen in the core, the generation of energy via the pp chain must stop. However, by now the core temperature has increased to the point that nuclear fusion continues to generate energy in a thick hydrogen-burning shell around a small, predominantly helium core. This effect can be seen in the luminosity curve in Fig. 13.3. Note that the luminosity remains close to zero throughout the inner 3% of the star's mass. At the same time, the temperature is nearly constant over the same region. That the helium core must be isothermal when the luminosity gradient is zero can be seen from the radiative temperature gradient, given by Eq. (10.68). Since $L_r \simeq 0$ over a finite region, $dT/dr \simeq 0$

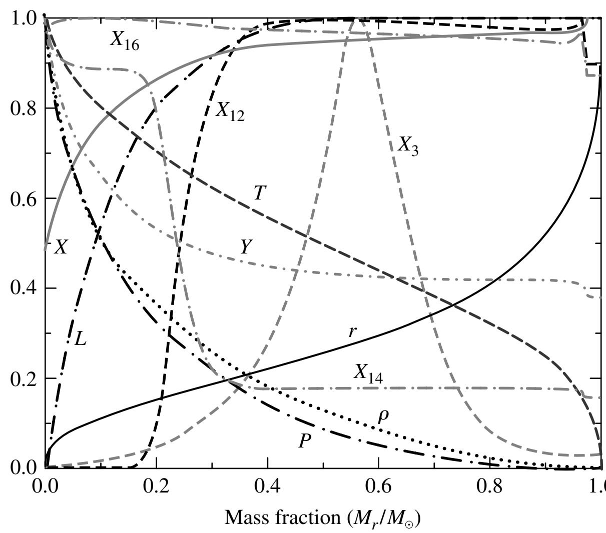


FIGURE 13.2 The interior structure of the present-day Sun (a 1 $M_\odot$ star), 4.57 Gyr after reaching the ZAMS. The model is located between points 1 and 2 in Fig. 13.1. The maximum ordinate values of the parameters are $r = 1.0\ R_\odot$, $L = 1.0\ L_\odot$, $T = 15.69 \times 10^6$ K, $\rho = 1.527 \times 10^5$ kg m^{-3}, $P = 2.342 \times 10^{16}$ N m^{-2}, $X = 0.73925$, $Y = 0.64046$, $X_3 = 3.19 \times 10^{-3}$, $X_{12} = 3.21 \times 10^{-3}$, $X_{14} = 5.45 \times 10^{-3}$, and $X_{16} = 9.08 \times 10^{-3}$. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, *555*, 990, 2001.)

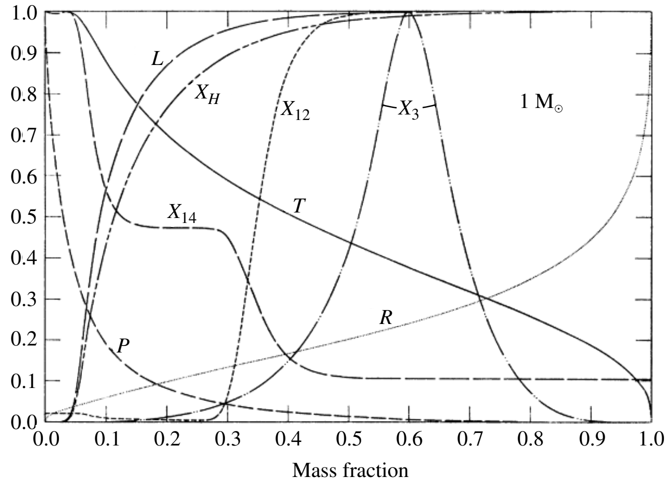


FIGURE 13.3 The interior structure of a 1 $M_\odot$ star near point 3 in Fig. 13.1, as described by the pioneering calculations of Icko Iben. Although specific values of quantities in modern models differ somewhat from those given here, state-of-the-art models do not significantly differ qualitatively from these calculations. The maximum ordinate values of the parameters for the Iben model are $R = 1.2681\ \mathrm{R}_\odot$, $P = 1.3146 \times 10^{17}\ \mathrm{N\ m^{-2}}$, $T = 19.097 \times 10^6$ K, $L = 2.1283\ \mathrm{L}_\odot$, $X_H = 0.708$, $X_3 = 5.15 \times 10^{-3}$, $X_{12} = 3.61 \times 10^{-3}$, and $X_{14} = 1.15 \times 10^{-2}$. The radius of the star is $1.3526\ \mathrm{R}_\odot$. (Figure adapted from Iben, *Ap. J.*, *47*, 624, 1967.)

and T is nearly constant. For an isothermal core to support the material above it in hydrostatic equilibrium, the required pressure gradient must be the result of a continuous increase in density as the center of the star is approached.

At this point, the luminosity being generated in the thick shell actually exceeds what was produced by the core during the phase of core hydrogen burning. As a result, the evolutionary track continues to rise beyond point 3 in Fig. 13.1, although not all of the energy generated reaches the surface; some of it goes into a slow expansion of the envelope. Consequently, the effective temperature begins to decrease slightly and the evolutionary track bends to the right. As the hydrogen-burning shell continues to consume its nuclear fuel, the ash from nuclear burning causes the isothermal helium core to grow in mass while the star moves farther to the red in the H–R diagram.

The Schönberg–Chandrasekhar Limit

This phase of evolution ends when the mass of the isothermal core has become too great and the core is no longer capable of supporting the material above it. The maximum fraction of a star's mass that can exist in an isothermal core and still support the overlying layers was first estimated by M. Schönberg and Chandrasekhar in 1942; it is given by

$$\boxed{\left(\frac{M_{ic}}{M}\right)_{\mathrm{SC}} \simeq 0.37 \left(\frac{\mu_{\mathrm{env}}}{\mu_{ic}}\right)^2,} \tag{13.1}$$

where μ_{env} and μ_{ic} are the mean molecular weights of the overlying envelope and the isothermal core, respectively. The **Schönberg–Chandrasekhar limit** is another consequence of the virial theorem. Based on the physical tools we have developed so far, an approximate form of this result can be obtained. The analysis is presented beginning on the facing page.

The maximum fraction of the mass of a star that can be contained in an isothermal core and still maintain hydrostatic equilibrium is a function of the mean molecular weights of the core and the envelope. When the mass of the isothermal helium core exceeds this limit, the core collapses on a Kelvin–Helmholtz timescale, and the star evolves very rapidly relative to the nuclear timescale of main-sequence evolution. This occurs at the points labeled 4 in Fig. 13.1. For stars below about 1.2 $\mathrm{M}_\odot$, this defines the end of the main-sequence phase. What happens next is the subject of Section 13.2.

Example 13.1.1. If a star is formed with the initial composition $X = 0.68$, $Y = 0.30$, and $Z = 0.02$, and if complete ionization is assumed at the core–envelope boundary, we find from Eq. (10.16) that $\mu_{\text{env}} \simeq 0.63$. Assuming that all of the hydrogen has been converted into helium in the isothermal core, $\mu_{ic} \simeq 1.34$. Therefore, from Eq. (13.1), the Schönberg–Chandrasekhar limit is

$$\left(\frac{M_{ic}}{M}\right)_{\text{SC}} \simeq 0.08.$$

The isothermal core will collapse if its mass exceeds 8% of the star's total mass.

The Degenerate Electron Gas

The mass of an isothermal core may exceed the Schönberg–Chandrasekhar limit if an additional source of pressure can be found to supplement the ideal gas pressure. This can occur if the electrons in the gas start to become **degenerate**. When the density of a gas becomes sufficiently high, the electrons in the gas are forced to occupy the lowest available energy levels. Since electrons are fermions and obey the Pauli exclusion principle (see Section 5.4), they cannot all occupy the same quantum state. Consequently, the electrons are stacked into progressively higher energy states, beginning with the ground state. In the case of *complete degeneracy*, the pressure of the gas is due entirely to the resultant nonthermal motions of the electrons, and therefore it becomes independent of the temperature of the gas.

If the electrons are nonrelativistic, the pressure of a completely degenerate electron gas is given by

$$P_e = K\rho^{5/3}, \tag{13.2}$$

where K is a constant.[3] If the degeneracy is only partial, some temperature dependence remains.[4] The isothermal core of a 1 $\mathrm{M}_\odot$ star between points 3 and 4 in Fig. 13.1 is partially degenerate; consequently, the core mass can reach approximately 13% of the entire mass

[3]Note that Eq. (13.2) is a polytropic equation of state with a polytropic index of $n = 1.5$; see page 334ff.
[4]The physics of degenerate gases will be discussed in more detail in Section 16.3.

of the star before it begins to collapse. Less massive stars exhibit even higher levels of degeneracy on the main sequence and may not exceed the Schönberg–Chandrasekhar limit at all before the next stage of nuclear burning commences.

Main-Sequence Evolution of Massive Stars

The evolution of more massive stars on the main sequence is similar to that of their lower-mass cousins with one important difference: the existence of a convective core. The convection zone continually mixes the material, keeping the core composition nearly homogeneous. This is because the timescale for convection, defined by the amount of time it takes a convective element to travel one mixing length (see Section 10.4), is much shorter than the nuclear timescale. For a 5 $M_\odot$ star, the central convection zone decreases somewhat in mass during core hydrogen burning, leaving behind a slight composition gradient. Moving up the main sequence, as the star evolves the convection zone in the core retreats more rapidly with increasing stellar mass, disappearing entirely before the hydrogen is exhausted for those stars with masses greater than about 10 $M_\odot$.

When the mass fraction of hydrogen reaches about $X = 0.05$ in the core of a 5 $M_\odot$ star (point 2 in Fig. 13.1), the entire star begins to contract. With the release of some gravitational potential energy, the luminosity increases slightly. Since the radius decreases, the effective temperature must also increase. For stars with masses greater than 1.2 $M_\odot$, this stage of overall contraction is defined to be the end of the main-sequence phase of evolution.

A Derivation of the Schönberg–Chandrasekhar limit

To estimate the Schönberg–Chandrasekhar limit, begin by dividing the equation of hydrostatic equilibrium (Eq. 10.6) by the equation of mass conservation (Eq. 10.7). This gives

$$\boxed{\frac{dP}{dM_r} = -\frac{GM_r}{4\pi r^4},} \tag{13.3}$$

which is just the condition of hydrostatic equilibrium, written with the interior mass as the independent variable.[5] Rewriting, Eq. (13.3) may be expressed as

$$4\pi r^3 \frac{dP}{dM_r} = -\frac{GM_r}{r}. \tag{13.4}$$

The left-hand side is just

$$4\pi r^3 \frac{dP}{dM_r} = \frac{d(4\pi r^3 P)}{dM_r} - 12\pi r^2 P \frac{dr}{dM_r} = \frac{d(4\pi r^3 P)}{dM_r} - \frac{3P}{\rho},$$

where Eq. (10.7) was used to obtain the last expression. Substituting back into Eq. (13.4) and integrating over the mass (M_{ic}) of the isothermal core, we have

$$\int_0^{M_{ic}} \frac{d(4\pi r^3 P)}{dM_r}\, dM_r - \int_0^{M_{ic}} \frac{3P}{\rho}\, dM_r = -\int_0^{M_{ic}} \frac{GM_r}{r}\, dM_r. \tag{13.5}$$

[5]This is called the Lagrangian form of the condition for hydrostatic equilibrium.

To evaluate Eq. (13.5), we will consider each term separately. The first term on the left-hand side is just

$$\int_0^{M_{ic}} \frac{d(4\pi r^3 P)}{dM_r}\, dM_r = 4\pi R_{ic}^3 P_{ic},$$

where R_{ic} and P_{ic} are the radius and the gas pressure at the *surface* of the isothermal core, respectively (note that $r = 0$ at $M_r = 0$).

The second term on the left-hand side of Eq. (13.5) can also be evaluated quickly by realizing that, from the ideal gas law,

$$\frac{P}{\rho} = \frac{kT_{ic}}{\mu_{ic} m_H},$$

where T_{ic} and μ_{ic} are the temperature and mean molecular weight throughout the isothermal core, respectively.[6] Thus

$$\int_0^{M_{ic}} \frac{3P}{\rho}\, dM_r = \frac{3M_{ic}kT_{ic}}{\mu_{ic} m_H} = 3N_{ic}kT_{ic} = 2K_{ic},$$

where

$$N_{ic} \equiv \frac{M_{ic}}{\mu_{ic} m_H}$$

is the number of gas particles in the core and

$$K_{ic} = \frac{3}{2} N_{ic} k T_{ic}$$

is the total thermal energy of the core, assuming an ideal monatomic gas.

The right-hand side of Eq. (13.5) is simply the gravitational potential energy of the core, or

$$-\int_0^{M_{ic}} \frac{GM_r}{r}\, dM_r = U_{ic}.$$

Substituting each term into Eq. (13.5), we find

$$4\pi R_{ic}^3 P_{ic} - 2K_{ic} = U_{ic}. \tag{13.6}$$

This expression should be compared with the form of the virial theorem developed in Section 2.4; see Eq. (2.46). If we had integrated from the center of the star to the surface, where $P \simeq 0$, we would have obtained our original version of the theorem. The difference lies in the nonzero pressure boundary condition. Thus Eq. (13.6) is a generalized form of the virial theorem for stellar interiors in hydrostatic equilibrium.

[6]The core is actually supported in part by electron degeneracy pressure, meaning that the ideal gas law is not strictly valid. For our purposes here, however, the assumption of an ideal gas gives reasonable results.

Next, from Eq. (10.22), the gravitational potential energy of the core may be approximated as

$$U_{ic} \sim -\frac{3}{5}\frac{GM_{ic}^2}{R_{ic}}.$$

Furthermore, the internal thermal energy of the core is just

$$K_{ic} = \frac{3M_{ic}kT_{ic}}{2\mu_{ic}m_H}.$$

Introducing these expressions into Eq. (13.6) and solving for the pressure at the surface of the isothermal core, we have

$$P_{ic} = \frac{3}{4\pi R_{ic}^3}\left(\frac{M_{ic}kT_{ic}}{\mu_{ic}m_H} - \frac{1}{5}\frac{GM_{ic}^2}{R_{ic}}\right). \tag{13.7}$$

Notice that there are two competing terms in Eq. (13.7); the first term is due to the thermal energy in the core and the second is due to gravitational effects. For specific values of T_{ic} and R_{ic}, as the core mass increases, the thermal energy tends to increase the pressure at the surface of the core while the gravitational term tends to decrease it. For some value of M_{ic}, P_{ic} is maximized, meaning that there exists an upper limit on how much pressure the isothermal core can exert in order to support the overlying envelope.

To determine when P_{ic} is a maximum, we must differentiate Eq. (13.7) with respect to M_{ic} and set the derivative equal to zero. It is left as an exercise to show that the radius of the isothermal core for which P_{ic} is a maximum is given by

$$R_{ic} = \frac{2}{5}\frac{GM_{ic}\mu_{ic}m_H}{kT_{ic}} \tag{13.8}$$

and that the maximum value of the surface pressure that can be produced by an isothermal core is given by

$$P_{ic,\max} = \frac{375}{64\pi}\frac{1}{G^3M_{ic}^2}\left(\frac{kT_{ic}}{\mu_{ic}m_H}\right)^4. \tag{13.9}$$

The important feature of Eq. (13.9) is that, as the core mass increases, the maximum pressure at the surface of the core *decreases*. At some point, it may no longer be possible for the core to support the overlying layers of the star's envelope. Clearly this critical condition must be related to the mass contained in the envelope and therefore to the total mass of the star.

To estimate the mass that can be supported by the isothermal core, we need to determine the pressure exerted on the core by the overlying envelope. In hydrostatic equilibrium, this pressure must not exceed the maximum possible pressure that may be supported by the isothermal core. To estimate the envelope pressure, we will start again with Eq. (13.3), this time integrating from the surface of the star to the surface of the isothermal core. Assuming for simplicity that the pressure at the surface of the star is zero,

$$\begin{aligned} P_{ic,\text{env}} &= \int_0^{P_{ic,\text{env}}} dP \\ &= -\int_M^{M_{ic}} \frac{GM_r}{4\pi r^4}\, dM_r \\ &\simeq -\frac{G}{8\pi\left\langle r^4\right\rangle}\left(M_{ic}^2 - M^2\right), \end{aligned}$$

where M is the total mass of the star and $\langle r^4\rangle$ is some average value of r^4 between the surface of the star of radius R and the surface of the core. Assuming that $M_{ic}^2 \ll M^2$, and making the crude approximation that $\langle r^4\rangle \sim R^4/2$, we have

$$P_{ic,\text{env}} \sim \frac{G}{4\pi}\frac{M^2}{R^4} \tag{13.10}$$

for the pressure at the core's surface due to the weight of the envelope.

The quantity R^4 can be written in terms of the mass of the star and the temperature of the isothermal core through the use of the ideal gas law,

$$T_{ic} = \frac{P_{ic,\text{env}}\mu_{\text{env}} m_H}{\rho_{ic,\text{env}} k}, \tag{13.11}$$

where μ_{env} is the mean molecular weight of the envelope and $\rho_{ic,\text{env}}$ is the gas density at the core–envelope interface. Making the rough estimate that

$$\rho_{ic,\text{env}} \sim \frac{M}{4\pi R^3/3}, \tag{13.12}$$

using Eq. (13.10), and solving for R, Eq. (13.11) gives

$$R \sim \frac{1}{3}\frac{GM}{T_{ic}}\frac{\mu_{\text{env}} m_H}{k}.$$

Substituting our solution for the radius of the envelope back into Eq. (13.10), we arrive at an expression for the pressure at the core–envelope interface due to the overlying envelope:

$$P_{ic,\text{env}} \sim \frac{81}{4\pi}\frac{1}{G^3 M^2}\left(\frac{kT_{ic}}{\mu_{\text{env}} m_H}\right)^4.$$

Note that $P_{ic,\text{env}}$ is independent of the mass of the isothermal core.

Finally, to estimate the Schönberg–Chandrasekhar limit, we set the maximum pressure of the isothermal core (Eq. 13.9) equal to the pressure needed to support the overlying envelope. This immediately simplifies to give

$$\frac{M_{ic}}{M} \sim 0.54\left(\frac{\mu_{\text{env}}}{\mu_{ic}}\right)^2.$$

Our result is only slightly larger than the one obtained originally by Schönberg and Chandrasekhar (Eq. 13.1).

13.2 ■ LATE STAGES OF STELLAR EVOLUTION

Following the completion of the main-sequence phase of stellar evolution, a complicated sequence of evolutionary stages occurs that may involve nuclear burning in the cores of stars together with nuclear burning in concentric mass shells. At various times, core burning and/or nuclear burning in a mass shell may cease, accompanied by a readjustment of the structure of the star. This readjustment may involve expansion or contraction of the core or envelope and the development of extended convection zones. As the final stages of evolution are approached, extensive mass loss from the surface also plays a critical role in determining the star's ultimate fate.

As examples of post-main-sequence stellar evolution, we will continue to explore changes in the structures over time of a *low-mass star* of 1 $M_\odot$ and an *intermediate-mass star* of 5 $M_\odot$. Detailed depictions of their evolutionary tracks in the H–R diagram are shown in Figs. 13.4 and 13.5, respectively.

Evolution Off the Main Sequence

As mentioned in Section 13.1, the end of the main-sequence phase of evolution occurs when hydrogen burning ceases in the core of the star (in Fig. 13.1 this corresponds to point 3 for the 1 $M_\odot$ star and point 2 for the 5 $M_\odot$ star). In the case of the 1 $M_\odot$ star, the core begins to contract while a thick hydrogen-burning shell continues to consume available fuel. With the rising temperature in the shell due to core contraction, the shell actually produces more energy than the core did on the main sequence, causing the luminosity to increase, the envelope to expand slightly, and the effective temperature to decrease.

The situation is somewhat different for the 5 $M_\odot$ star, however. Rather than a thick hydrogen-burning shell immediately producing energy with the cessation of hydrogen burning in the core, the entire star participates in an overall contraction on a Kelvin–Helmholtz timescale. This contraction phase releases gravitational potential energy, causing the luminosity to increase slightly, the radius of the star to decrease, and the effective temperature to increase (corresponding to the evolution between points 2 and 3 in Fig. 13.1). Eventually the temperature outside the helium core increases sufficiently to cause a thick shell of hydrogen to burn (point 3 in Fig. 13.1). At this point the interior chemical composition of the 5 $M_\odot$ star resembles that of Fig. 13.6. Because the ignition of the shell is quite rapid, the overlying envelope is forced to expand slightly, absorbing some of the energy released by the shell. As a result, the luminosity decreases momentarily and the effective temperature drops, as can be seen in both Figs. 13.1 and 13.5. A sketch of the star's structure at this point is given in Fig. 13.7.

The Subgiant Branch

For both low- and intermediate-mass stars, as the shell continues to consume the hydrogen that is available at the base of the star's envelope, the helium core steadily increases in mass

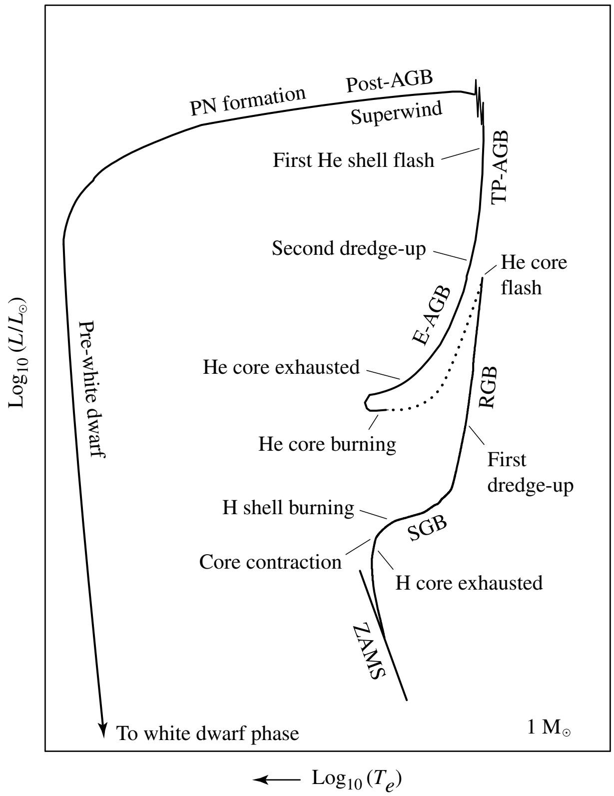


FIGURE 13.4 A schematic diagram of the evolution of a low-mass star of 1 $M_\odot$ from the zero-age main sequence to the formation of a white dwarf star (see Section 16.1). The dotted phase of evolution represents rapid evolution following the helium core flash. The various phases of evolution are labeled as follows: Zero-Age-Main-Sequence (ZAMS), Sub-Giant Branch (SGB), Red Giant Branch (RGB), Early Asymptotic Giant Branch (E-AGB), Thermal Pulse Asymptotic Giant Branch (TP-AGB), Post-Asymptotic Giant Branch (Post-AGB), Planetary Nebula formation (PN formation), and Pre-white dwarf phase leading to white dwarf phase.

and becomes nearly isothermal. At points 4 in Fig. 13.1, the Schönberg–Chandrasekhar limit is reached and the core begins to contract rapidly, causing the evolution to proceed on the much faster Kelvin–Helmholtz timescale. The gravitational energy released by the rapidly contracting core again causes the envelope of the star to expand and the effective temperature cools, resulting in redward evolution on the H–R diagram. This phase of evolution is known as the **subgiant branch** (SGB).

As the core contracts, a nonzero temperature gradient is soon re-established because of the release of gravitational potential energy. At the same time, the temperature and density of the hydrogen-burning shell increase, and, although the shell begins to narrow significantly, the rate at which energy is generated by the shell increases rapidly. Once again the stellar envelope expands, absorbing some of the energy produced by the shell

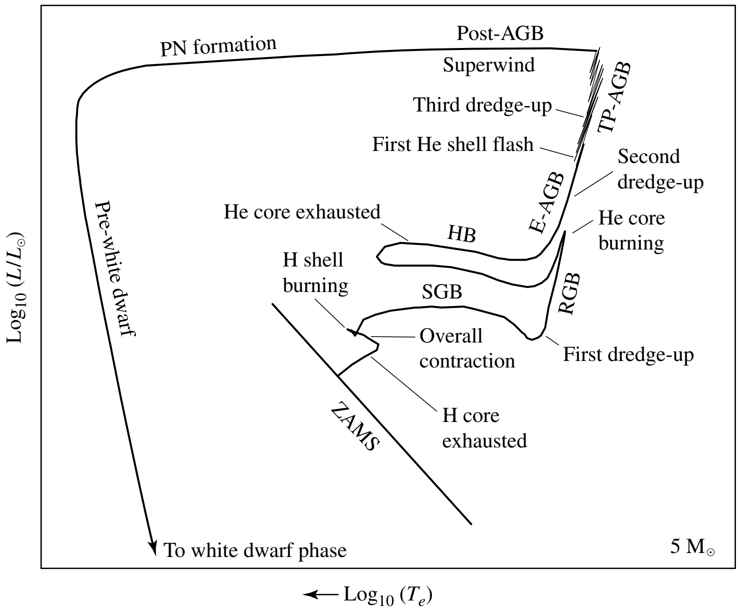


FIGURE 13.5 A schematic diagram of the evolution of an intermediate-mass star of 5 $M_\odot$ from the zero-age main sequence to the formation of a white dwarf star (see Section 16.1). The diagram is labeled according to Fig. 13.4 with the addition of the Horizontal Branch (HB).

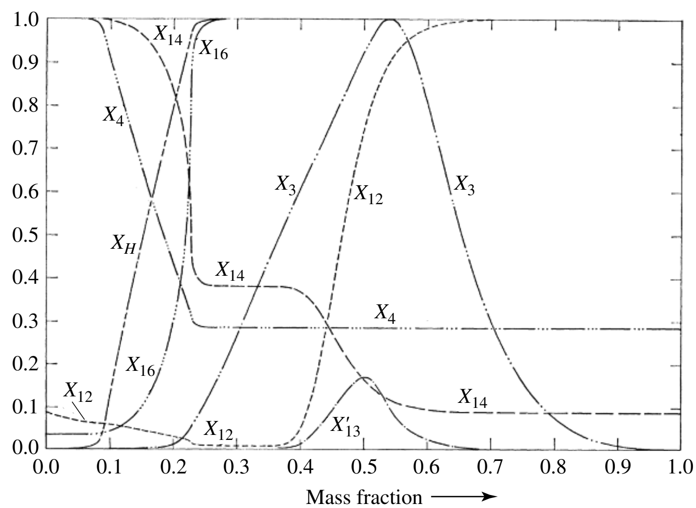


FIGURE 13.6 The chemical composition as a function of interior mass fraction for a 5 $M_\odot$ star during the phase of overall contraction, following the main-sequence phase of core hydrogen burning. The maximum mass fractions of the indicated species are $X_H = 0.708$, $X_3 = 1.296 \times 10^{-4}$ ($^3_2$He), $X_4 = 0.9762$ ($^4_2$He), $X_{12} = 3.61 \times 10^{-3}$ ($^{12}_{6}$C), $X'_{13} = 3.61 \times 10^{-3}$ ($^{13}_{6}$C), $X_{14} = 0.0145$ ($^{14}_{7}$N), and $X_{16} = 0.01080$ ($^{16}_{8}$O). (Figure adapted from Iben, *Ap. J.*, *143*, 483, 1966.)

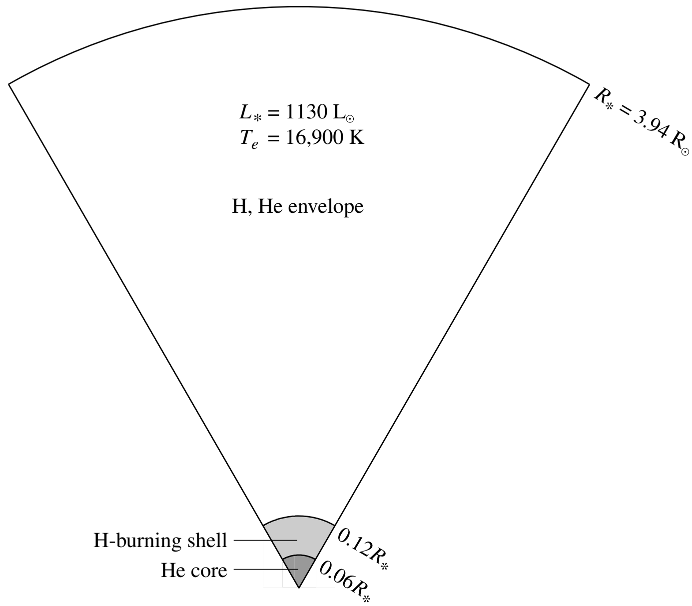


FIGURE 13.7 A 5 $M_\odot$ star with a helium core and a hydrogen-burning shell shortly after shell ignition (point 3 in Fig. 13.1). (Data from Iben, *Ap. J.*, *143*, 483, 1966.)

before the energy reaches the surface. For the 5 $M_\odot$ star, in a situation analogous to thick hydrogen shell burning immediately following overall contraction, the expanding envelope actually absorbs enough energy for a time to cause the luminosity to decrease slightly before recovering (point 5 in Fig. 13.1).

The Red Giant Branch

With the expansion of the stellar envelope and the decrease in effective temperature, the photospheric opacity increases due to the additional contribution of the H^- ion. The result is that a convection zone develops near the surface for both low- and intermediate-mass stars. As the evolution continues toward points 5 in Fig. 13.1, the base of the convection zone extends deep into the interior of the star. With the nearly adiabatic temperature gradient associated with convection throughout much of the stellar interior, and the efficiency with which the energy is transported to the surface, the star begins to rise rapidly upward along the **red giant branch** (RGB) of the H-R diagram. This path is essentially the same one followed by pre-main-sequence stars descending the Hayashi track prior to the onset of core hydrogen burning.

As the star climbs the RGB, its convection zone deepens until the base reaches down into regions where the chemical composition has been modified by nuclear processes. In particular, because of its rather large nuclear reaction cross section, lithium burns via collisions with protons at relatively cool temperatures (greater than about 2.7×10^6 K). This means that because of the evolution of the star to this point, lithium has become nearly depleted over most of the interior of the star (the inner 98% of the mass for the 5 $M_\odot$ star).[7] At the same time, nuclear processing has increased the mass fraction of ${}^3_2$He over the middle third of the star (see, for example, Figs. 13.3 and 13.6) as well as altered the abundance ratios of the various species in the CNO cycle. When the surface convection zone encounters this chemically modified region, the processed material becomes mixed with the material above it. The effect is observable changes in the composition of the photosphere; the amount of lithium at the surface will decrease and the amount of ${}^3_2$He will increase. At the same time, convection transports ${}^{12}_6$C inward and ${}^{14}_7$N outward, decreasing the observable ratio of X_{12}/X_{14}. Other abundance ratios such as X'_{13}/X_{12} will also be modified. This transport of materials from the deep interior to the surface is referred to as the **first dredge-up** phase. Nature has provided us with an opportunity to directly observe the products of nuclear reactions deep within stellar interiors. These observable changes in surface composition provide an important test of the predictions of stellar evolution theory.

The Red Giant Tip

At the tip of the RGB (points 6 in Fig. 13.1) the central temperature and density (1.3×10^8 K and 7.7×10^6 kg m^{-3} for the 5 $M_\odot$ star) have finally become high enough that quantum-mechanical tunneling becomes effective through the Coulomb barrier acting between ${}^4_2$He nuclei, allowing the triple alpha process to begin. Some of the resulting ${}^{12}_6$C is further processed into ${}^{16}_8$O as well.

With the onset of a new and strongly temperature-dependent source of energy (see Eqs. 10.60 and 10.61), the core expands. Although the hydrogen-burning shell remains the dominant source of the star's luminosity, the expansion of the core pushes the hydrogen-burning shell outward, cooling it and causing the rate of energy output of the shell to decrease somewhat. The result is an abrupt decrease in the luminosity of the star. At the same time, the envelope contracts and the effective temperature begins to increase again.

The Helium Core Flash

An interesting difference arises at this point between the evolution of stars with masses greater than about 1.8 $M_\odot$ and those that have masses less than 1.8 $M_\odot$. For stars of lower mass, as the helium core continues to collapse during evolution up to the tip of the red giant branch, the core becomes strongly electron-degenerate. Furthermore, significant neutrino losses from the core of the star prior to reaching the tip of the RGB result in a negative temperature gradient near the center (i.e., a temperature inversion develops); the core is actually refrigerated somewhat because of the energy that is carried away by the easily escaping neutrinos! When the temperature and density become high enough to initiate the triple alpha process (approximately 10^8 K and 10^7 kg m^{-3}, respectively), the ensuing energy release is almost explosive. The ignition of helium burning occurs initially in a shell around the center of the star, but the entire core quickly becomes involved and the temperature inversion is lifted. The luminosity generated by the helium-burning core reaches 10^{11} $L_\odot$, comparable to that of an entire galaxy! However, this tremendous energy release lasts for

[7]Recall from Section 11.1 that the surface abundance of lithium is also lower than expected in the present-day Sun.

only a few seconds, and most of the energy never even reaches the surface. Instead, it is absorbed by the overlying layers of the envelope, possibly causing some mass to be lost from the surface of the star. This short-lived phase of evolution of low-mass stars is referred to as the **helium core flash**. The origin of the explosive energy release is in the very weak temperature dependence of electron degeneracy pressure and the strong temperature dependence of the triple alpha process. The energy generated must first go into "lifting" the degeneracy. Only after this occurs can the energy go into thermal (kinetic) energy required to expand the core, which decreases the density, lowers the temperature, and slows the reaction rate.

It is because of the very rapid pace of the helium core flash that stellar evolution calculations of low-mass stars are often terminated at that point. Given the dramatic changes occuring deep in the interior of the star, it is very difficult to follow the evolution adequately; very small time steps are required to model the evolution, meaning that a great deal of computer time is needed to follow a star through the helium core flash (in fact, the star evolves *much faster* than the computer can model it). This is why the evolutionary tracks of stars with masses of 1, 1.25, and 1.5 $M_\odot$ are not followed past points 6 in Fig. 13.1. This is also why the annotated evolutionary track in Fig. 13.4 immediately following the red giant tip is indicated by a dotted line; the evolution is extremely rapid and the computations are resumed when quiescent helium core burning and hydrogen shell burning are established in the star.

The Horizontal Branch

For both low- and intermediate-mass stars, as the envelope of the model contracts following the red giant tip, the increasing compression of the hydrogen-burning shell eventually causes the energy output of the shell, and the overall energy output of the stars, to begin to rise again. With the associated increase in effective temperature, the deep convection zone in the envelope rises toward the surface, while at the same time, a convective core develops. The appearance of a convective core is due to the high temperature sensitivity of the triple alpha process (just as the convective core of an upper-main-sequence star arises because of the temperature dependence of the CNO cycle). This generally horizontal evolution is the blueward portion of the **horizontal branch** (HB) loop. The blueward portion of the HB is essentially the helium-burning analog of the hydrogen-burning main sequence, but with a much shorter timescale.

When the evolution of the star reaches its most blueward point (point 8 in Fig. 13.1 for the 5 $M_\odot$ star), the mean molecular weight of the core has increased to the point that the core begins to contract, accompanied by the expansion and cooling of the star's envelope. Shortly after beginning the redward portion of the HB loop, the core helium is exhausted, having been converted to carbon and oxygen. Again the redward evolution proceeds rapidly as the inert CO core contracts, much like the rapid evolution across the SGB following the extinction of core hydrogen burning.

During their passage along the horizontal branch, many stars develop instabilities in their outer envelopes, leading to periodic pulsations that are readily observable as variations in luminosity, temperature, radius, and surface radial velocity. Since these oscillations depend sensitively on the internal structure of the star, stellar pulsations provide yet another test of stellar structure theory.[8]

With the increase in core temperature associated with its contraction, a thick helium-burning shell develops outside the CO core. As the core continues to contract, the helium-burning shell narrows and strengthens, forcing the material above the shell to expand and cool. This results in a temporary turn-off of the hydrogen-burning shell.

Along with the contraction of the helium-exhausted core, neutrino production increases to the point that the core cools a bit. As a consequence of the increasing central density and decreasing temperature, electron degeneracy pressure becomes an important component of the total pressure in the carbon–oxygen core.

The Early Asymptotic Giant Branch

The next phase of the evolution illustrated in Figs. 13.4 and 13.5 is very similar to the evolution following the exhaustion of the hydrogen-burning cores. When the redward evolution reaches the Hayashi track, the evolutionary track bends upward along a path referred to as the **asymptotic giant branch** (AGB). (The AGB is so named because the evolutionary track approaches the line of the RGB asymptotically from the left. The AGB may be thought of as the helium-burning-shell analog to the hydrogen-burning-shell RGB.) At this point in its evolution the core temperature of the 5 $M_\odot$ star is approximately 2×10^8 K, and its density is on the order of 10^9 kg m^{-3}. A diagram of an ***early* AGB** (E-AGB) star with two shell sources is shown in Fig. 13.8. Although two shell sources are depicted, it is the helium-burning shell that dominates the energy output during the E-AGB; the hydrogen-burning shell is nearly inactive at this point. Note that the diagram is not to scale; in order to visualize the structure from the hydrogen-burning shell inward, that region was enlarged by a factor of 100 relative to the surface of the star.

The expanding envelope initially absorbs much of the energy produced by the helium-burning shell. As the effective temperature continues to decrease, the convective envelope deepens again, this time extending downward to the chemical discontinuity between the hydrogen-rich outer layer and the helium-rich region above the helium-burning shell. The mixing that results during this **second dredge-up** phase increases the helium and nitrogen content of the envelope. (The increase in nitrogen is due to the previous conversion of carbon and oxygen into nitrogen in the intershell region.)

The Thermal-Pulse Asymptotic Giant Branch

Near the upper portion of the AGB (labeled TP-AGB in Figs. 13.4 and 13.5 for ***thermal-pulse* AGB**), the dormant hydrogen-burning shell eventually reignites and again dominates the energy output of the star. However, during this phase of evolution, the narrowing helium-burning shell begins to turn on and off quasi-periodically. These intermittent **helium shell flashes** occur because the hydrogen-burning shell is dumping helium ash onto the helium layer below. As the mass of the helium layer increases, its base becomes slightly degenerate. Then, when the temperature at the base of the helium shell increases sufficiently, a helium

[8]Pulsating variable stars will be discussed in greater detail in Chapter 14.

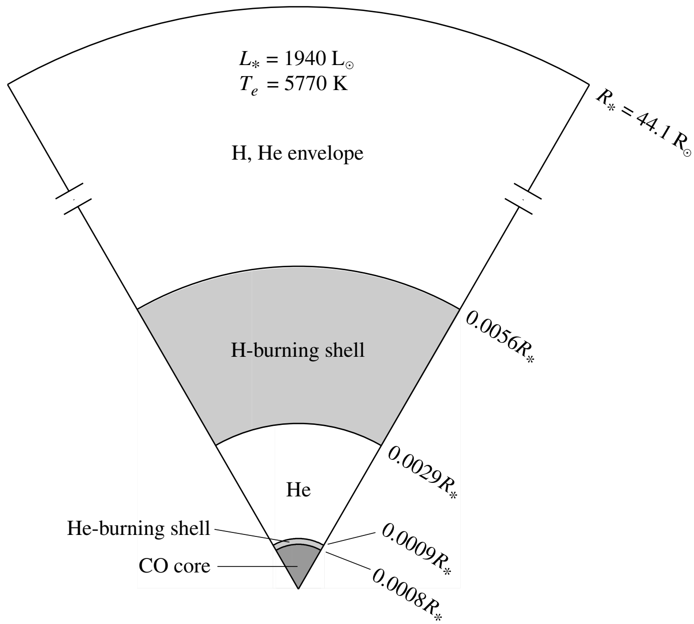


FIGURE 13.8 A 5 $M_\odot$ star on the early asymptotic giant branch with a carbon–oxygen core and hydrogen- and helium-burning shells. Note that relative to the surface radius, the scale of the shells and core has been increased by a factor of 100 for clarity. (Data from Iben, *Ap. J.*, *143*, 483, 1966.)

shell flash occurs, analogous to the earlier helium core flashes of low-mass stars (although much less energetic). This drives the hydrogen-burning shell outward, causing it to cool and turn off for a time. Eventually the burning in the helium shell diminishes, the hydrogen-burning shell recovers, and the process repeats. The period between pulses is a function of the mass of the star, ranging from thousands of years for stars near 5 $M_\odot$ to hundreds of thousands of years for low-mass stars (0.6 $M_\odot$), with the pulse amplitude growing with each successive event; see Fig. 13.9. This phase of periodic activity in the deep interior of the star is evident in abrupt changes in luminosity at the surface (see the TP-AGB phases in Figs. 13.4 and 13.5).

Details of thermal pulses for a 7 $M_\odot$ star are shown in Fig. 13.10. Following a helium shell flash, the luminosity arising from the hydrogen-burning shell drops appreciably while the energy output from the helium-burning shell increases. This is because the hydrogen-burning shell is pushed outward, causing it to cool. Since the hydrogen-burning shell is responsible for most of the energy output of the star, the star's luminosity abruptly decreases when a helium shell flash occurs. At the same time, the radius of the surface of the star also decreases and the star's effective temperature increases. After a period of time, the energy output of the helium shell diminishes when the degeneracy is lifted, the hydrogen-burning shell moves deeper into the star, and the hydrogen-burning shell once again dominates the star's total energy output. As a result, the surface radius, luminosity, and effective temperature relax back to near their pre-flash values. It is important to note, however, that

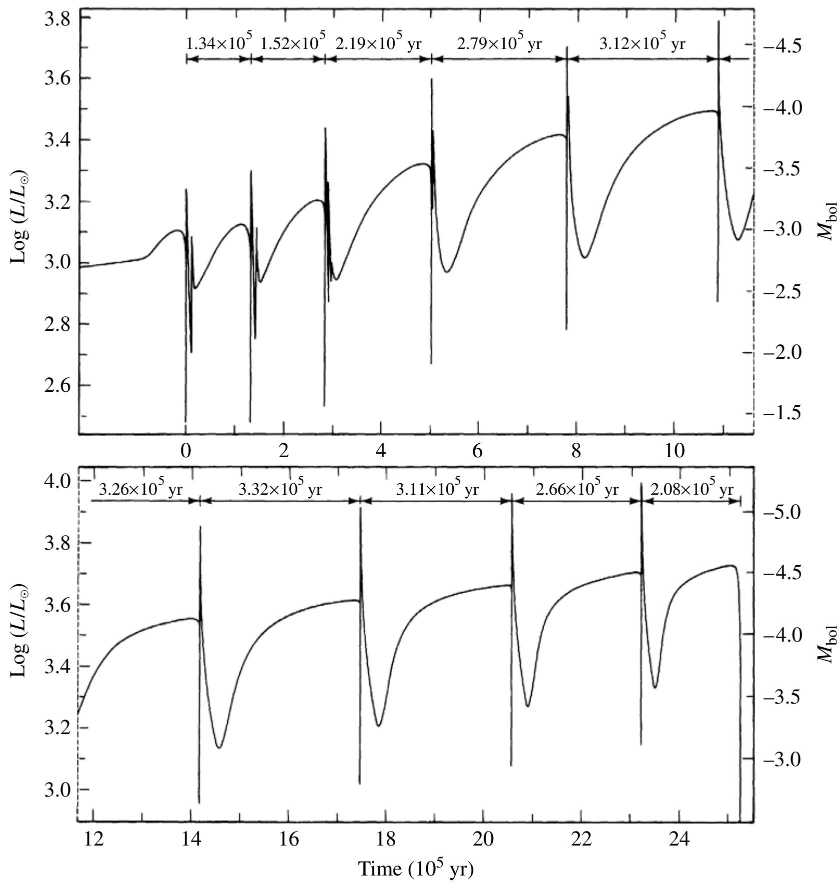


FIGURE 13.9 The surface luminosity as a function of time for a 0.6 $M_\odot$ stellar model that is undergoing helium shell flashes on the TP-AGB. (Figure adapted from Iben, *Ap. J.*, *260*, 821, 1982.)

the overall evolutionary track of the star is toward greater luminosity and lower effective temperature throughout the TP-AGB.

A class of pulsating variable stars known as **long-period variables** (LPVs) are AGB stars. (LPVs have pulsation periods of 100 to 700 days and include the subclass of **Mira variable stars**.) It has been suggested that the structural changes arising from shell flashes could cause observable changes in the periods of some of these stars, providing another possible test of stellar evolution theory. In fact, several Miras (e.g., W Dra, R Aql, and R Hya) have been observed to be undergoing relatively rapid period changes.

Third Dredge-Up and Carbon Stars

Because of the sudden increase in energy flux from the helium-burning shell during a flash episode, a convection zone is established between the helium-burning shell and the hydrogen-burning shell. At the same time, the depth of the envelope convection zone

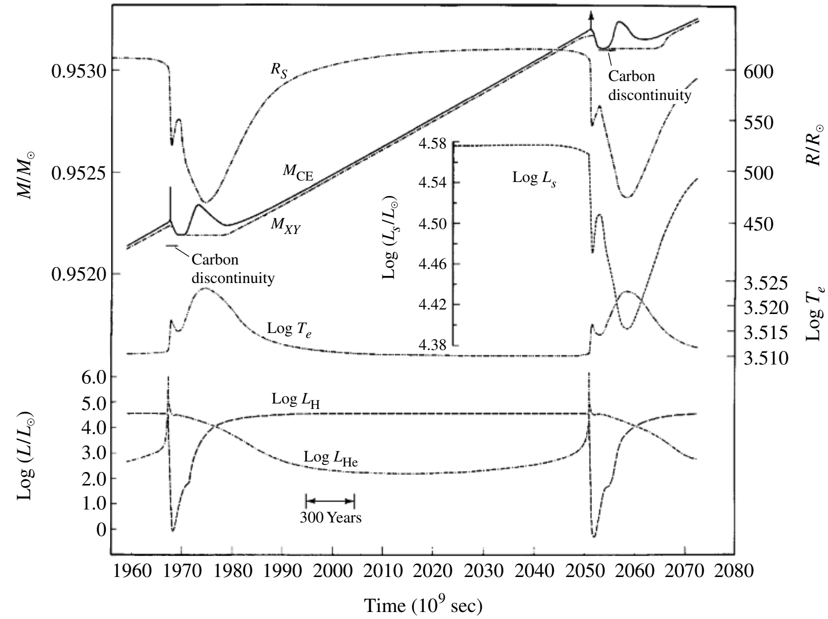


FIGURE 13.10 Time-dependent changes in the properties of a 7 $M_\odot$ AGB star produced by helium shell flashes on the TP-AGB. The quantities shown are the surface radius (R_S), the interior mass fractions of the base of the convective envelope (M_{CE}) and the hydrogen–helium discontinuity (M_{XY}), the star's luminosity and effective temperature (L_s and T_e, respectively), and the luminosities of the hydrogen- and helium-burning shells (L_H and L_{He}, respectively). (Figure adapted from Iben, *Ap. J.*, *196*, 525, 1975.)

increases with the pulse strength of the flashes. For stars that are massive enough ($M > 2\ M_\odot$), the convection zones will merge and eventually extend down into regions where carbon has been synthesized. In the region between the hydrogen- and helium-burning shells, the abundance of carbon exceeds that of oxygen by a factor of five to ten. This is in sharp contrast to the general excess of oxygen over carbon in the atmospheres of most stars. During this **third dredge-up** phase, the carbon-rich material is brought to the surface, decreasing the ratio of oxygen to carbon. If there are multiple third dredge-up events arising from repeated helium shell flashes, the oxygen-rich spectrum of a star will transform over time to a carbon-rich spectrum. This appears to explain the difference, observed spectroscopically, between oxygen-rich giants where the number density of oxygen atoms in the atmosphere exceeds the number density of carbon atoms ($N_O > N_C$) and carbon-rich giants ($N_C > N_O$) called **carbon stars**.

Carbon stars are designated with a special **C spectral type** (overlapping the traditional K and M types). These stars are distinguished by an abundance of carbon-rich molecules in their atmospheres, such as SiC, rather than SiO of typical M stars. This occurs because carbon monoxide (CO) is a very tightly bound molecule. If the atmosphere of the star contains more oxygen than carbon, the carbon is almost completely tied up in CO, leaving oxygen to form additional molecules. Conversely, if the atmosphere contains more carbon than oxygen, the oxygen is tied up in CO, allowing carbon to form in molecules.

Intermediate between the M and C spectral types are the **S spectral type** stars. These stars show ZrO lines in their atmospheres, having replaced the TiO lines of M stars. S stars have almost identical abundances of carbon and oxygen in their atmospheres. S and C spectral types represent additions to the spectral classification scheme presented in Table 8.1.

Of particular interest in the atmospheres of evolved TP-AGB stars is the presence of technetium (Tc), an element with no stable isotopes. In particular, $^{99}_{43}$Tc is the most abundant isotope of technetium found in the atmospheres of TP-AGB stars, yet it has a half-life of only 200,000 yr. The existence of technetium in S and C stars strongly suggests that the isotope must have been formed very recently in the star's history and dredged up to the surface from the deep interior.

s-Process Nucleosynthesis

Technetium-99 is one of a host of isotopes formed by the slow capture of neutrons by existing nuclei. Numerous nuclear reactions, such as carbon burning (Eq. 10.66) and oxygen burning (Eq. 10.67) release neutrons. Since neutrons do not have an electric charge, they can easily collide with nuclei (there is no Coulomb barrier to tunnel through). If the flux of neutrons is not too great, radioactive nuclei produced by the absorption of stray neutrons have time to decay into other nuclei before they absorb another neutron. $^{99}_{43}$Tc is one product of this *slow* ***s*-process** nucleosynthesis.

Mass Loss and AGB Evolution

AGB stars are known to lose mass at a rapid rate, sometimes as high as $\dot{M} \sim 10^{-4}\ \mathrm{M}_\odot\ \mathrm{yr}^{-1}$. The effective temperatures of these stars are also quite cool (around 3000 K). As a result, dust grains form in the expelled matter. Since silicate grains tend to form in an environment rich in oxygen, and graphite grains will form in a carbon-rich environment, the composition of the ISM may be related to the relative numbers of carbon- and oxygen-rich stars. Observations of ultraviolet extinction curves in the Milky Way and the Large and Small Magellanic Clouds[9] support the idea that mass loss from these stars does, in fact, help enrich the ISM.

As evolution up the AGB continues, what happens next is strongly dependent on the original mass of the star and the amount of mass loss experienced by that star during its lifetime. It appears that the final evolutionary behavior of stars can be separated into two basic groups: those with ZAMS masses above about 8 $\mathrm{M}_\odot$ and those with masses below this value. The distinction between the two mass regimes is based on whether or not the core of the star will undergo significant further nuclear burning. In the remainder of this section, we will consider the final evolution of stars with initial masses less than 8 $\mathrm{M}_\odot$, leaving the ultimate evolution of more massive stars to Chapter 15.

As stars with initial masses below 8 $\mathrm{M}_\odot$ continue to evolve up the AGB, the helium-burning shell converts more and more of the helium into carbon and then into oxygen, increasing the mass of the carbon–oxygen core. At the same time, the core continues to contract slowly, causing its central density to increase. Depending on the star's mass, neutrino energy losses may decrease the central temperature somewhat during this phase. In any event, the densities in the core become large enough that electron degeneracy pressure

[9]The LMC and the SMC are small satellite galaxies of the Milky Way, visible in the southern hemisphere.

begins to dominate. This situation is very similar to the development of an electron-degenerate helium core in a low-mass star during its rise up the red giant branch.

For stars with ZAMS masses less than about 4 $M_\odot$, the carbon–oxygen core will never become large enough and hot enough to ignite nuclear burning. On the other hand, *if the important contribution of mass loss is ignored* for stars between 4 $M_\odot$ and 8 $M_\odot$, theory suggests that the C–O core would reach a sufficiently large mass that it could no longer remain in hydrostatic equilibrium, even with the assistance of pressure from the degenerate electron gas. The outcome of this situation is catastrophic core collapse. The maximum value of 1.4 $M_\odot$ for a completely degenerate core is known as the **Chandrasekhar limit**.[10]

However, as has already been mentioned, observations of AGB stars do show enormous mass loss rates. When these mass loss rates are included in evolution calculations on the AGB, the situation described in the last paragraph does not actually occur. Instead, for stars between 4 $M_\odot$ and 8 $M_\odot$, mass loss prevents catastrophic core collapse. Instead of collapse, these stars experience additional nucleosynthesis in their cores, leading to core compositions of oxygen, neon, and magnesium (ONeMg cores) and masses remaining below the Chandrasekhar limit of 1.4 $M_\odot$. Nuclear reactions capable of producing this composition were given in Eqs. (10.66) and (10.67). In addition, reactions such as

$$^{22}_{10}\mathrm{Ne} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{25}_{12}\mathrm{Mg} + \mathrm{n}$$

$$^{22}_{10}\mathrm{Ne} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{26}_{12}\mathrm{Mg} + \gamma$$

also affect the composition of these cores.

Unfortunately, our understanding of the mechanism(s) that cause this mass loss is poor. Some astronomers have suggested that the mass loss may be linked to the helium shell flashes or perhaps to the periodic envelope pulsations of LPVs. Other proposed mechanisms stem from the high luminosities and low surface gravities of these stars, coupled with radiation pressure on the dust grains, dragging the gas with them. Whatever the cause, its influence on the evolution of AGB stars is significant.

As one might expect, the rate of mass loss accelerates with time because the luminosity and radius are increasing while the mass is decreasing during continued evolution up the AGB. The decreasing mass and increasing radius of the star imply that the surface gravity is also decreasing, and the surface material is becoming progressively less tightly bound. Consequently, mass loss becomes increasingly more important as AGB evolution continues.

In the latest stages of evolution on the AGB, a **superwind** develops with $\dot{M} \sim 10^{-4}\ M_\odot\ \mathrm{yr}^{-1}$. Whether shell flashes, envelope pulsations, or some other mechanism is the reason, the observed high mass loss rates seem to be responsible for the existence of a class of objects known as **OH/IR sources**. These objects appear to be stars shrouded in optically thick dust clouds that radiate their energy primarily in the infrared part of the electromagnetic spectrum.

The OH part of the OH/IR designation is due to the detection of OH molecules, which can be seen via their **maser emission**.[11] A maser is the molecular analog of a laser; electrons

[10] The Chandrasekhar limit plays a critical role in the formation of the final products of stellar evolution, namely white dwarfs, neutron stars, and black holes. The physics of the Chandrasekhar limit will be discussed in some detail in Section 16.4.

[11] The term *maser* is an acronym for *m*icrowave *a*mplification by *s*timulated *e*mission of *r*adiation.

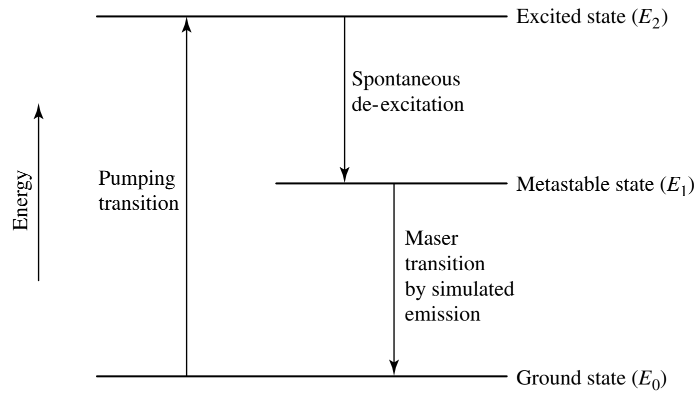


FIGURE 13.11 A schematic diagram of a hypothetical three-level maser. The intermediate energy level is a relatively long-lived metastable state. A transition from the metastable state to the lowest energy level can occur through stimulated emission by a photon of energy equal to the energy difference between the two states ($E_\gamma = h\nu = E_1 - E_0$).

are "pumped up" from a lower energy level into a higher, long-lived metastable energy state. The electron then makes a downward transition back to a lower state when it is stimulated by a photon with an energy equal to the difference in energies between the two states. The original photon and the emitted photon will travel in the same direction and will be in phase with each other; hence the amplification of radiation. A schematic energy level diagram of a hypothetical three-level maser is depicted in Fig. 13.11.

Post-Asymptotic Giant Branch

As the cloud around the OH/IR source continues to expand, it eventually becomes optically thin, exposing the central star, which characteristically exhibits the spectrum of an F or G supergiant. At this point in the evolution of our 1 and 5 $M_\odot$ stars (Figs. 13.4 and 13.5, respectively), the evolutionary tracks have turned blueward, leaving the TP-AGB and moving nearly horizontally across the H–R diagram as **post-AGB** stars. During the ensuing final phase of mass loss, the remainder of the star's envelope is expelled, revealing the cinders produced by its long history of nuclear reactions. With only a very thin layer of material remaining above them, the hydrogen- and helium-burning shells are extinguished, and the luminosity of the star drops rapidly. The hot central object, now revealed, will cool to become a **white dwarf star**, which is essentially the old red giant's degenerate C–O core (or ONeMg core in the case of the more massive stars), surrounded by a thin layer of residual hydrogen and helium. This important class of stars, the end products of the evolution of stars with initial main-sequence masses less than 8 $M_\odot$, will be discussed in Chapter 16.

The last stages of evolution of a 0.6 $M_\odot$ model are depicted in Fig. 13.12. The position of the star on the H–R diagram at the start of each flash episode is indicated by a number next to the evolutionary track (eleven pulses in all), with the resulting excursions in luminosity and effective temperature indicated for pulses 7, 9, and 10. It is after the tenth pulse that the star leaves the AGB, ejecting its envelope during the nearly constant luminosity path

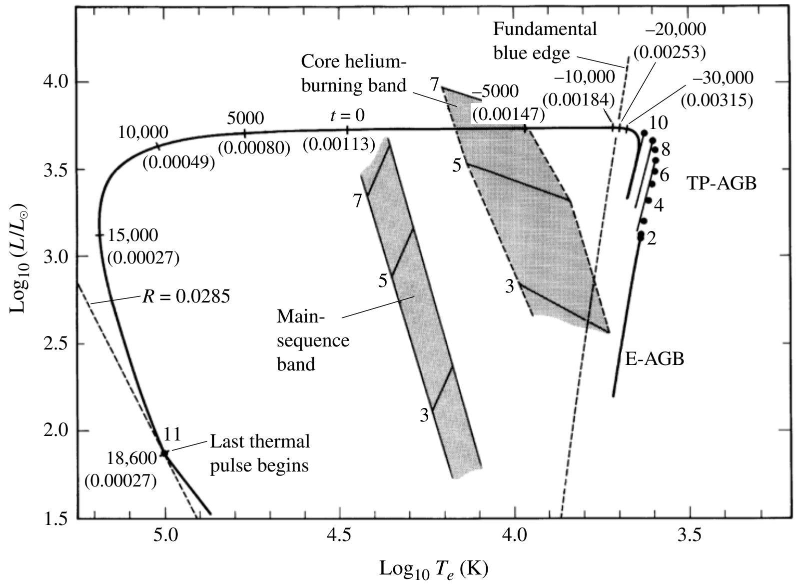


FIGURE 13.12 The AGB and post-AGB evolution of a 0.6 M$_\odot$ star undergoing mass loss. The initial composition of the model is $X = 0.749$, $Y = 0.25$, and $Z = 0.001$. The main-sequence and horizontal branches of 3, 5, and 7 M$_\odot$ stars are shown for reference. Details of the figure are discussed in the body of the text. (Figure adapted from Iben, *Ap. J.*, *260*, 821, 1982.)

across the H–R diagram. The amount of mass remaining in the hydrogen-rich envelope is indicated in parentheses along the evolutionary track (in M$_\odot$). Also indicated is the amount of time before (negative) or after (positive) the point when the star's effective temperature was 30,000 K (the time is measured in years). Following the eleventh helium shell flash, the star finally loses the last remnants of its envelope and becomes a white dwarf of radius 0.0285 R$_\odot$.[12]

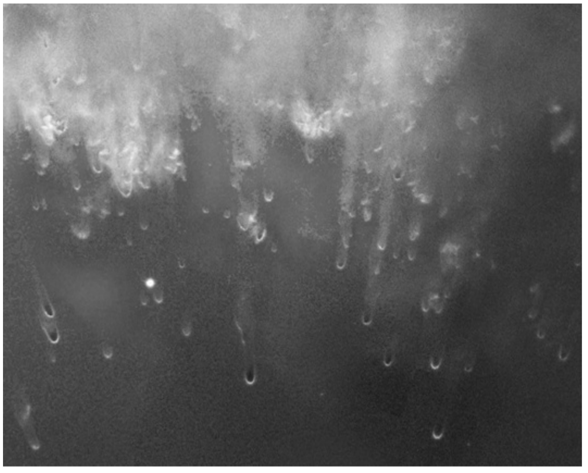
Planetary Nebulae

The expanding shell of gas around a white dwarf progenitor is called a **planetary nebula**. Examples of planetary nebulae are shown in Figs. 13.13–13.15. These beautiful, glowing clouds of gas were given this name in the nineteenth century because, when viewed through a small telescope, they look somewhat like giant gaseous planets.

A planetary nebula owes its appearance to the ultraviolet light emitted by the hot, condensed central star. The ultraviolet photons are absorbed by the gas in the nebula, causing the atoms to become excited or ionized. When the electrons cascade back down to lower

[12]The line labeled "Fundamental blue edge" corresponds to the high-temperature limit for fundamental mode pulsations of a class of variable stars known as **RR Lyraes**. This important class of objects will be discussed extensively in Chapter 14.

(a)



(b)

FIGURE 13.13 (a) The Helix nebula (NGC 7293) is one of the closest planetary nebulae to Earth, 213 pc away in the constellation of Aquarius. Its angular diameter in the sky is about 16 arcmin, roughly one-half the angular size of the full moon. The pre-white dwarf star is visible at the center of the nebula. [Credit: NASA, ESA, C.R. O'Dell (Vanderbilt University), M. Meixner, and P. McCullough.] (b) A close-up of "cometary knots" in the Helix nebula. The central star is located beyond the bottom of the picture. [Credit: NASA, NOAO, ESA, the Hubble Helix Nebula Team, M. Meixner (STScI), and T. A. Rector (NRAO).]

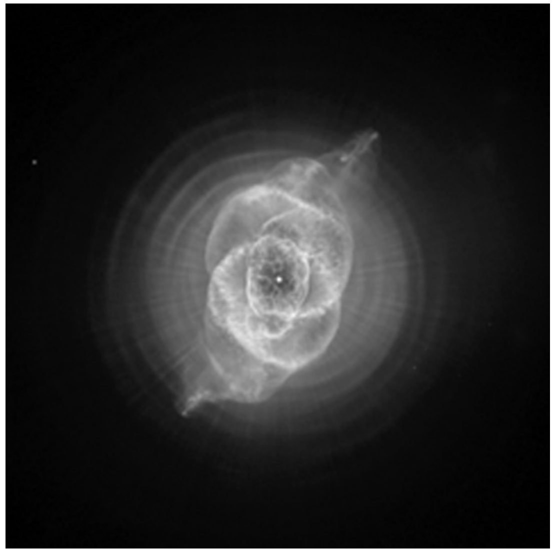


FIGURE 13.14 NGC 6543 (the "Cat's Eye") is a planetary nebula in Draco, 900 pc away. The complex structure may be due to high-speed jets and the presence of a companion star, making NGC 6543 part of a binary star system. The jets are clearly visible in the upper right-hand and lower left-hand portions of the image. Note the central star in the image. [Credit: NASA, ESA, HEIC, and the Hubble Heritage Team (STScI/AURA). Acknowledgment: R. Corradi (Isaac Newton Group of Telescopes, Spain) and Z. Tsvetanov (NASA).]

energy levels, photons are emitted whose wavelengths are in the visible portion of the electromagnetic spectrum. As a result, the cloud appears to glow in visible light.[13]

The bluish-green coloration of many planetary nebulae is due to the 500.68-nm and 495.89-nm forbidden lines of [O III] (forbidden lines of [O II] and [Ne III] are also common), and the reddish coloration comes from ionized hydrogen and nitrogen. Characteristic temperatures of these objects are in the range of the ionization temperature of hydrogen, 10^4 K.

With the advent of high-resolution images of planetary nebulae obtained by telescopes such as the Hubble Space Telescope, astronomers have come to realize that the morphologies of planetary nebulae are often much more complex than might have been expected of a spherically symmetric parent TP-AGB star. Some planetaries, like the Helix nebula in Fig. 13.13(a), look as though they have a ringlike structure. This is because gas is ejected preferentially along the equator of the star due to the presence of angular momentum, and our viewing angle is down the star's rotation axis. Suggestions for the surprising array of structures include varying viewing angles, multiple ejections of material from the stellar surface, the presence of one or more companion stars, and magnetic fields.

[13]This process is reminiscent of the creation of H II regions around newly formed O and B main-sequence stars, discussed in Section 12.3.

(a)

(b)

FIGURE 13.15 Examples of two "butterfly" planetary nebulae. (a) M2-9 is a bipolar planetary nebula 800 pc distant in Ophiucus. [Credit: Bruce Balick (University of Washington), Vincent Icke (Leiden University, The Netherlands), Garrelt Mellema (Stockholm University), and NASA.] (b) Menzel 3 (Mz 3), in Norma, is also known as the Ant nebula. Outflow velocities of 1000 km s^{-1} are much greater than for any other similar object. [Credit: NASA, ESA, and the Hubble Heritage Team (STScI/AURA). Acknowledgment: R. Sahai (Jet Propulsion Lab) and B. Balick (University of Washington).]

Significant detail is also evident at smaller scales. Figure 13.13(b) shows so-called cometary knots that are pointed radially away from the central star in the Helix nebula. These clumps of material have dark cores with luminous cusps on the sides facing the star.

The expansion velocities of planetaries, as measured by Doppler-shifted spectral lines, show that the gas is typically moving away from the central stars with speeds of between 10 and 30 km s^{-1}, although much greater speeds have been measured, as in the case of Mz 3 [Fig. 13.15(b)]. Combined with characteristic length scales of around 0.3 pc, their estimated ages are on the order of 10,000 years. After only about 50,000 years, a planetary nebula will dissipate into the ISM. Compared with the entire lifetime of a star, the phase of planetary nebula ejection is fleeting indeed.

Despite their short lifetimes, roughly 1500 planetary nebulae are known to exist in the Milky Way Galaxy. Given the fact that we are unable to observe the entire galaxy from

Earth, it is estimated that the number of planetaries is probably close to 15,000. If, on average, each planetary contains about 0.5 $M_\odot$ of material, the ISM is being enriched at the rate of roughly one solar mass per year through this process.

13.3 ■ STELLAR CLUSTERS

Over the past two chapters we have seen a story develop that depicts the lives of stars. They are formed from the ISM, only to return most of that material to the ISM through stellar winds, by the ejection of planetary nebulae, or via supernova explosions (to be discussed in Section 15.3). The matter that is given back, however, has been enriched with heavier elements that were produced through the various sequences of nuclear reactions governing a star's life. As a result, when the next generation of stars is formed, it possesses higher concentrations of these heavy elements than did its ancestors. This cyclic process of star formation, death, and rebirth is evident in the variations in composition between stars.

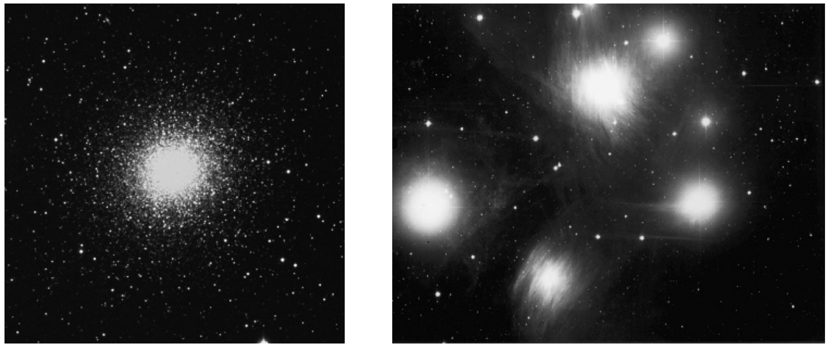
Population I, II, and III Stars

The universe began with the Big Bang 13.7 billion years ago. At that time hydrogen and helium were essentially the only elements produced by the nucleosynthesis that occurred during the initial fireball. Consequently, the first stars to form did so with virtually no metal content; $Z = 0$. The next generation of stars that formed were extremely **metal-poor**, having very low but non-zero values of Z. Each succeeding generation of star production resulted in higher and higher proportions of heavier elements, leading to metal-rich stars for which Z may reach values as high as 0.03. The (thus far hypothetical) original stars that formed immediately after the Big Bang are referred to as **Population III** stars, metal-poor stars with $Z \gtrsim 0$ are referred to as **Population II**, and metal-rich stars are called **Population I**.

The classifications of Population II and Population I are due originally to their identifications with kinematically distinct groups of stars within our Galaxy. Population I stars have velocities relative to the Sun that are low compared to Population II stars. Furthermore, Population I stars are found predominantly in the disk of the Milky Way, while Population II stars can be found well above or below the disk. It was only later that astronomers realized that these two groups of stars differed chemically as well. Not only do populations tell us something about evolution, but the kinematic characteristics, positions, and compositions of Population I and Population II stars also provide us with a great deal of information about the formation and evolution of the Milky Way Galaxy.

Globular Clusters and Galactic (Open) Clusters

Recall from Section 12.2 that during the collapse of a molecular cloud, **stellar clusters** can form, ranging in size from tens of stars to hundreds of thousands of stars. Every member of a given cluster formed from the same cloud, they all formed with essentially identical compositions, and they all formed within a relatively short period of time. Thus, excluding such effects as rotation, magnetic fields, and membership in a binary star system, the Vogt–Russell theorem suggests that the differences in evolutionary states between the various stars in the cluster are due solely to their initial masses.



(a) (b)

FIGURE 13.16 (a) M13, the great globular cluster in Hercules, is located approximately 7000 pc from Earth. (From the Digitized Sky Survey at STScI. Courtesy of Palomar/Caltech, the National Geographic Society, and the Space Telescope Science Institute.) (b) The Pleiades is a galactic cluster found in the constellation of Taurus, at a distance of 130 pc. (Courtesy of the National Optical Astronomical Observatories.)

Extreme Population II clusters formed when the Galaxy was very young, making them some of the oldest objects in the Milky Way. They also contain the largest number of members. Figure 13.16(a) shows M13, one such **globular cluster**, located in the constellation of Hercules. Population I clusters, such as the Pleiades [Fig. 13.16(b)], tend to be smaller and younger. These smaller clusters are called alternately **galactic clusters** or **open clusters**.

Spectroscopic Parallax

The H–R diagrams of clusters can be constructed in a self-consistent way without knowledge of the exact distances to them. Since the dimensions of a typical cluster are small relative to its distance from Earth, little error is introduced by assuming that each member of the cluster has the same distance modulus. As a result, plotting the apparent magnitude rather than the absolute magnitude only amounts to shifting the position of each star in the diagram vertically by the same amount. By matching the observational main sequence of the cluster to a main sequence calibrated in absolute magnitude, the distance modulus of the cluster can be determined, giving the cluster's distance from the observer. This method of distance determination is known as **spectroscopic parallax** (the method is also often referred to as **main-sequence fitting**).

Color–Magnitude Diagrams

Rather than attempting to determine the effective temperature of every member of a cluster by undertaking a detailed spectral line analysis of each star (which would be a major project for a globular cluster, even assuming that the stars were bright enough to get good spectra), it is much faster to determine their color indices ($B - V$). With knowledge of the apparent magnitude and the color index of each star, a **color–magnitude diagram** can be constructed. Color–magnitude diagrams for M3 (a globular cluster) and h and χ Persei (a *double* galactic cluster) are shown in Figs. 13.17 and 13.18, respectively.

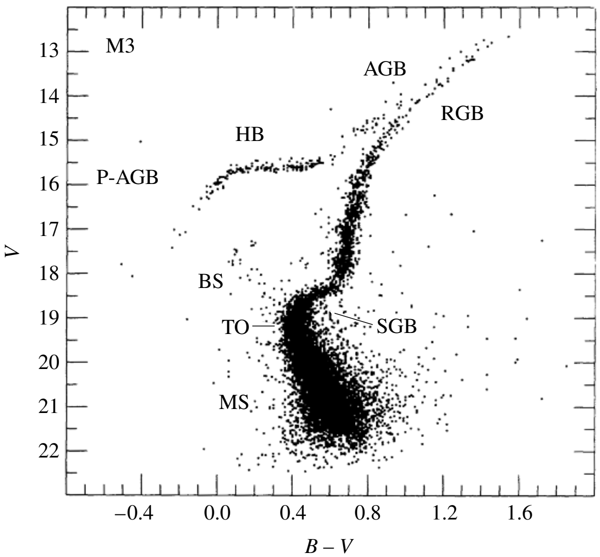


FIGURE 13.17 A color–magnitude diagram for M3, an old globular cluster. The major phases of stellar evolution are indicated: main sequence (MS); blue stragglers (BS); the main-sequence turn-off point (TO); the subgiant branch of hydrogen shell burning (SGB); the red giant branch along the Hayashi track, prior to helium core burning (RGB); the horizontal branch during helium core burning (HB); the asymptotic giant branch during hydrogen and helium shell burning (AGB); post-AGB evolution proceeding to the white dwarf phase (P-AGB). (Figure adapted from Renzini and Fusi Pecci, *Annu. Rev. Astron. Astrophys.*, *26*, 199, 1988. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 26, ©1988 by Annual Reviews Inc.)

Isochrones and Cluster Ages

Clusters, and their associated color–magnitude diagrams, offer nearly ideal tests of many aspects of stellar evolution theory. By computing the evolutionary tracks of stars of various masses, all having the same composition as the cluster, it is possible to plot the position of each evolving model on the H–R diagram when the model reaches the age of the cluster. (The curve connecting these positions is known as an **isochrone**.) The relative number of stars at each location on the isochrone depends on the number of stars in each mass range within the cluster (the initial mass function; see Fig. 12.12), combined with the different rates of evolution during each phase. Therefore, star counts in a color–magnitude diagram can shed light on the timescales involved in stellar evolution.

As the cluster ages, beginning with the initial collapse of the molecular cloud, the most massive and least abundant stars will arrive on the main sequence first, evolving rapidly. Before the lowest-mass stars have even reached the main sequence, the most massive ones have already evolved into the red giant region, perhaps even undergoing supernova explosions. These disparate rates of evolution can be seen by comparing Figs. 12.11 and 13.1 for pre-main-sequence and post-main-sequence evolution, respectively, together with their associated tables.

Since core hydrogen-burning lifetimes are inversely related to mass, continued evolution of the cluster means that the main-sequence **turn-off point**, defined as the point where stars

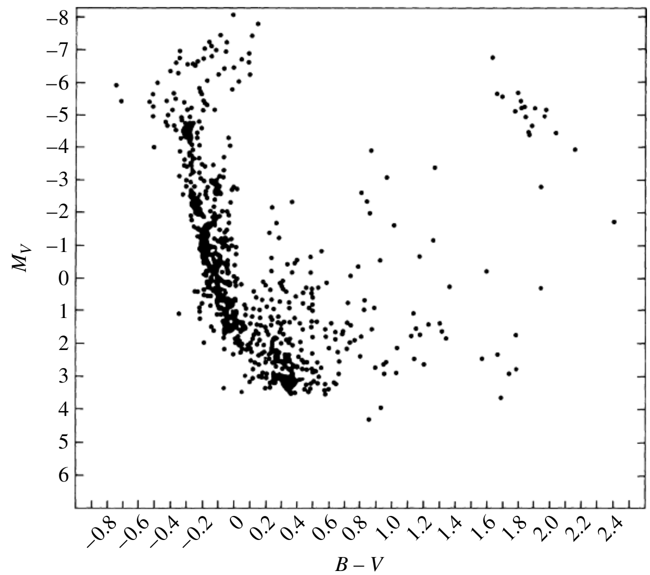


FIGURE 13.18 A color–magnitude diagram for the young double galactic cluster, h and χ Persei. Note that the most massive stars are pulling away from the main sequence while the low-mass stars in the middle of the diagram are still contracting onto the main sequence. Red giants are present in the upper right-hand corner of the diagram. (Figure adapted from Wildey, *Ap. J. Suppl.*, *8*, 439, 1964.)

in the cluster are currently leaving the main sequence, becomes redder and less luminous with time. Consequently, it is possible to estimate the age of a cluster by the location of the uppermost point of its main sequence. This fundamental technique is an important tool for determining ages of stars, clusters, our Milky Way Galaxy, and other galaxies with observable clusters, and even for establishing a lower limit on the age of the universe itself. A composite color–magnitude diagram of a number of clusters is shown in Fig. 13.19. Labeled vertically on the right-hand side is the age of the cluster corresponding to the location of the main-sequence turn-off point.

The Hertzsprung Gap

Another consequence of varying timescales can be seen in the color–magnitude diagram of h and χ Persei (Fig. 13.18). Apparent are red giants, together with low-mass pre-main-sequence stars. Also evident in the diagram is the complete absence of stars between the massive ones that are just leaving the main sequence and the few in the red giant region. It is unlikely that this represents an incomplete survey, since these stars are the brightest members of the cluster. Rather, it points out the very rapid evolution that occurs just after leaving the main sequence. This feature, known as the **Hertzsprung gap**, is a common characteristic of the color–magnitude diagrams of young, galactic clusters. The existence of the Hertzsprung gap is due to evolution on a Kelvin–Helmholtz timescale across the SGB, following the point when the hydrogen-depleted core exceeds the Schönberg–Chandrasekhar limit.

Notice in Fig. 13.19 that the cluster M67 does not show the existence of the Hertzsprung gap; the same can be said of M3 (Fig. 13.17). Recall that below about 1.25 $M_\odot$, the rapid

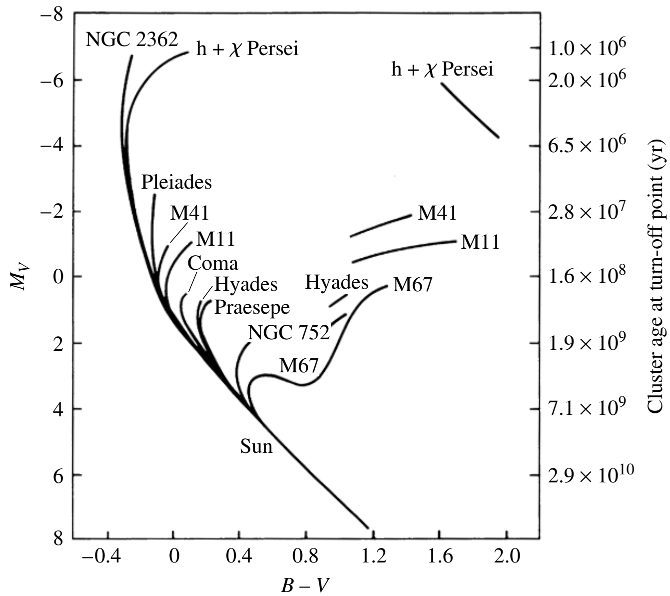


FIGURE 13.19 A composite color–magnitude diagram for a set of Population I galactic clusters. The absolute visual magnitude is indicated on the left-hand vertical axis, and the age of the cluster, based on the location of its turn-off point, is labeled on the right-hand side. (Figure adapted from an original diagram by A. Sandage.)

contraction phase related to the Schönberg–Chandrasekhar limit is much less pronounced. As a result, color–magnitude diagrams of old globular clusters with turn-off points near or less than 1 $M_\odot$ have continuous distributions of stars leading to the red giant region.

Relatively Few AGB and Post-AGB Stars

Close inspection of Fig. 13.17 also shows that a relatively small number of stars exist on the asymptotic giant branch and only a few stars are to be found in the region labeled P-AGB (post-asymptotic giant branch). This is just a consequence of the very rapid pace of evolution during this phase of heavy mass loss that leads directly to the formation of white dwarfs.

Blue Stragglers

It should be pointed out that a group of stars, known as **blue stragglers**, can be found above the turn-off point of M3. Although our understanding of these stars is incomplete, it appears that their tardiness in leaving the main sequence is due to some unusual aspect of their evolution. The most likely scenarios appear to be mass exchange with a binary star companion,[14] or collisions between two stars, extending the star's main-sequence lifetime.

[14]Mass exchange between close binaries is the subject of Chapter 18.

A Work in Progress

The successful comparisons between theory and observation that are provided by stellar clusters give strong support to the idea that our picture of stellar evolution is fairly complete, although perhaps in need of some fine-tuning. Continued refinements in stellar opacities, revisions in nuclear reaction cross sections, and much-needed improvements in the treatment of convection will probably lead to even better agreement with observations. However, much fundamental work remains to be done as well, such as developing a better understanding of the effects of mass loss, rotation, magnetic fields, and the presence of a close companion.

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PROBLEMS

13.1 **(a)** For a 5 $M_\odot$ star, use the data in Table 13.1 associated with Fig. 13.1 to construct a table that expresses the evolutionary times between points 2 and 3, points 3 and 4, and so on, as a percentage of the lifetime of the star on the main sequence between points 1 and 2.

(b) How long does it take a 5 $M_\odot$ star to cross the Hertzsprung gap relative to its main-sequence lifetime?

(c) How long does the 5 $M_\odot$ star spend on the blueward portion of the horizontal branch relative to its main-sequence lifetime?

(d) How long does the 5 $M_\odot$ star spend on the redward portion of the horizontal branch relative to its main-sequence lifetime?

13.2 Estimate the Kelvin–Helmholtz timescale for a 5 $M_\odot$ star on the subgiant branch and compare your result with the amount of time the star spends between points 4 and 5 in Fig. 13.1.

13.3 **(a)** Beginning with Eq. (13.7), show that the radius of the isothermal core for which the gas pressure is a maximum is given by Eq. (13.8). Recall that this solution assumes that the gas in the core is ideal and monatomic.

(b) From your results in part (a), show that the maximum pressure at the surface of the isothermal core is given by Eq. (13.9).

13.4 During the first dredge-up phase of a 5 $M_\odot$ star, would you expect the composition ratio X'_{13}/X_{12} to increase or decrease? Explain your reasoning. *Hint:* You may find Fig. 13.6 helpful.

13.5 Use Eq. (10.27) to show that the ignition of the triple alpha process at the tip of the red giant branch ought to occur at more than 10^8 K.

13.6 In an attempt to identify the important components of AGB mass loss, various researchers have proposed parameterizations of the mass loss rate that are based on fitting observed rates for a specified set of stars with some general equation that includes measurable quantities associated with the stars in the sample. One of the most popular, developed by D. Reimers, is given by

$$\dot{M} = -4 \times 10^{-13} \eta \frac{L}{gR} \, \mathrm{M}_\odot \, \mathrm{yr}^{-1}, \tag{13.13}$$

where L, g, and R are the luminosity, surface gravity, and radius of the star, respectively (all in solar units; $g_\odot = 274$ m s^{-2}). η is a *free parameter* whose value is expected to be near unity. Note that the minus sign has been explicitly included here, indicating that the mass of the star is decreasing.

(a) Explain qualitatively why L, g, and R enter Eq. (13.13) in the way they do.

(b) Estimate the mass loss rate of a 1 $M_\odot$ AGB star that has a luminosity of 7000 $L_\odot$ and a temperature of 3000 K.